

Captura de Diagramas Esquemáticos:

Vista general de la herramienta y del flujo de diseño a través de un ejemplo sencillo

Basado en materiales elaborados para Electratraining (www.electratraining.org) por:

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Agenda

- **Altium Designer Environment**
- Drawing Schematics
- PCB Editor

Elements of the Environment

The image shows the Altium Designer (18.1.9) interface with several callouts pointing to specific features:

- Menus, toolbars, shortcuts:** Points to the top menu bar (File, Edit, View, Project, Place, Design, Tools, Simulate, Reports, Window, Help) and the toolbar below it.
- Document tabs:** Points to the tabs at the top of the workspace, showing 'MiniPC.PcbDoc' and 'miniPCIE_Connector.SchDoc'.
- Main design window:** Points to the central workspace area displaying a schematic diagram of a miniPCIE connector.
- Properties panel:** Points to the 'Properties' panel on the right side of the interface.
- User selected panels:** Points to the 'Selection Filter' and 'General' sections within the Properties panel.
- Panel access:** Points to the 'Panel' button at the bottom right of the Properties panel.

The schematic diagram in the main design window shows a miniPCIE connector with various pins and components. The Properties panel on the right shows the 'General' section with units set to 'mils' and a visible grid of '100mil'.

Setup (license and updates)

The screenshot shows the Altium Designer (18.1.9) interface with the License Management window open. The window has three tabs: 'License Management', 'Extensions & Updates', and 'Updates'. The 'License Management' tab is active, showing the user is signed in as José F. Toledo. Below this, there are links for 'Sign out', 'Forgotten password?', 'Account preferences', and 'SUPPORTcenter'. The 'Available Licenses' section shows a table of licenses. A callout bubble labeled 'Access' points to the window's title bar. Another callout bubble labeled 'Updates' points to the 'Updates' tab. A third callout bubble labeled 'License setup' points to the 'License Management' tab.

License Management

License Management Extensions & Updates Updates

Altium Account - José F. Toledo is now signed in

Sign out Forgotten password? Account preferences SUPPORTcenter

Available Licenses - Licensed to [REDACTED] License subscription valid to 28-may

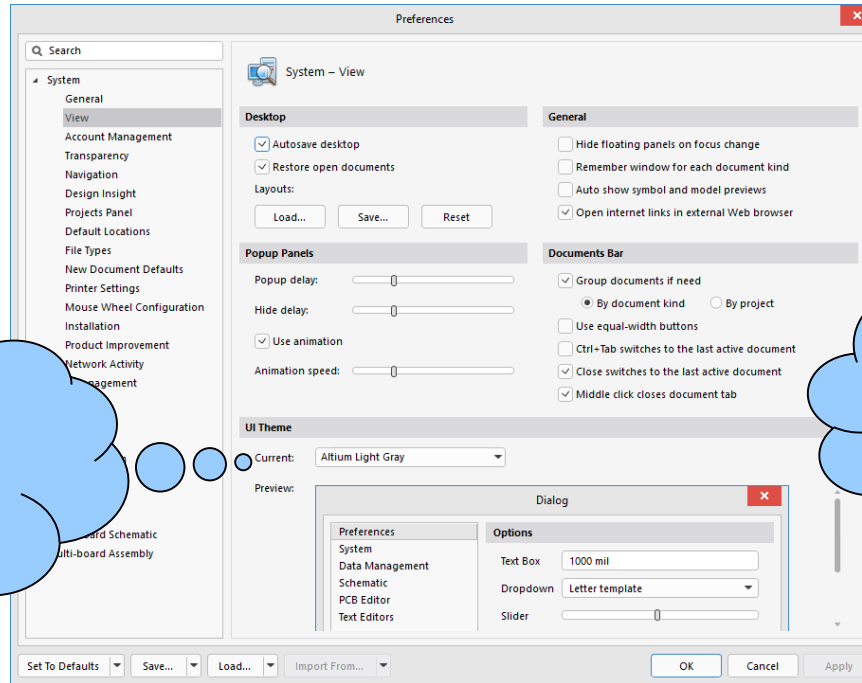
Product Name	Activation Code	Used	Assigned Seat Count	Expiry	Status	Subscription Status
Altium Designer	[REDACTED]	-	0/1	Infinite		Valid to 14-may.-2019
Altium Designer	[REDACTED]	-	1/1	Infinite	No seats left	Valid to 14-may.-2019
Altium Designer	[REDACTED]	-	0/1	Infinite		Valid to 23-jun.-2019
Altium Designer	[REDACTED]	-	0/1	Infinite	OK	Valid to 23-jun.-2019

Use Save standalone license file Extensions and updates
Roam Add standalone license file How to use an on-demand license
Release Setup private license server How to use a standalone license
Refresh Reconnect to private license server How to use a private server license

You are using one license.

Preferences

- Centralized preferences dialog – All tools are configured here
 - Preferences can be exported/imported to/from a file or a server

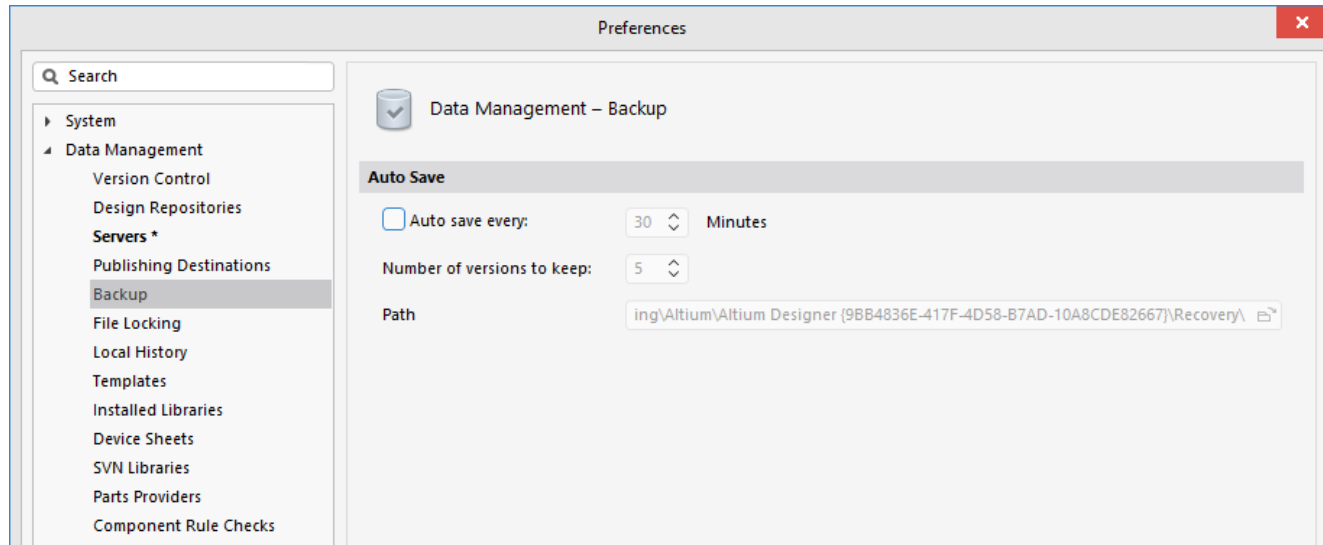


Select Dark or Light theme for the rest of the course here !!!

Screenshots will have a white background from now on!

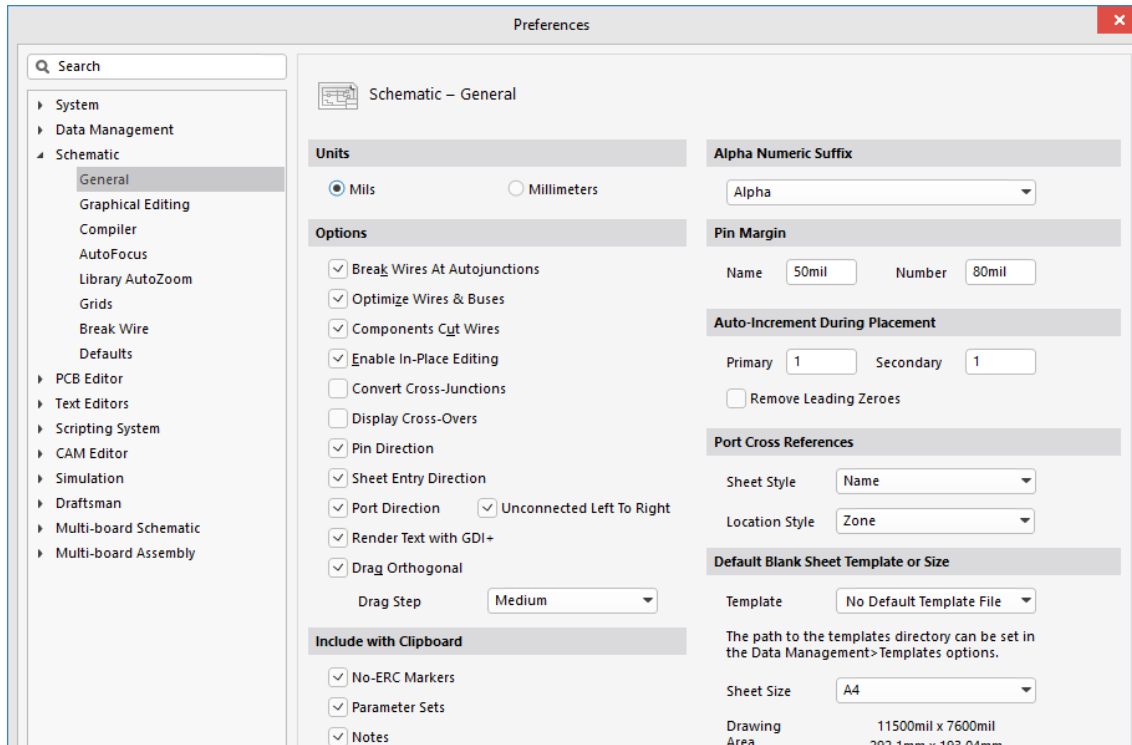
Preferences: Auto save

- **Auto Save** feature saves a copy of all currently open files that have been modified, into the specified location, at the specified time interval
 - Up to 10 versions of each file can be saved
 - Configured in **Preferences ➔ Data Management ➔ Backup**



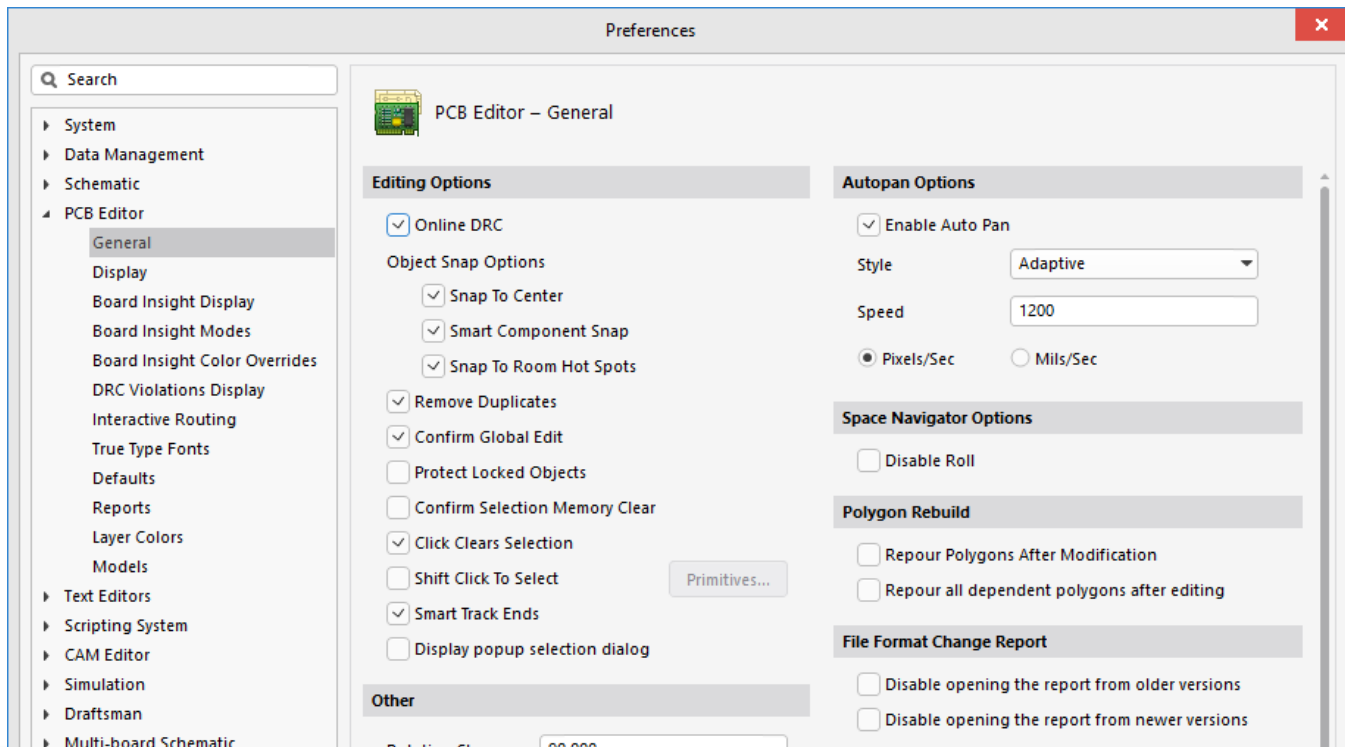
Preferences: Schematics

- Schematic ➔ General: Altium is a highly customizable tool



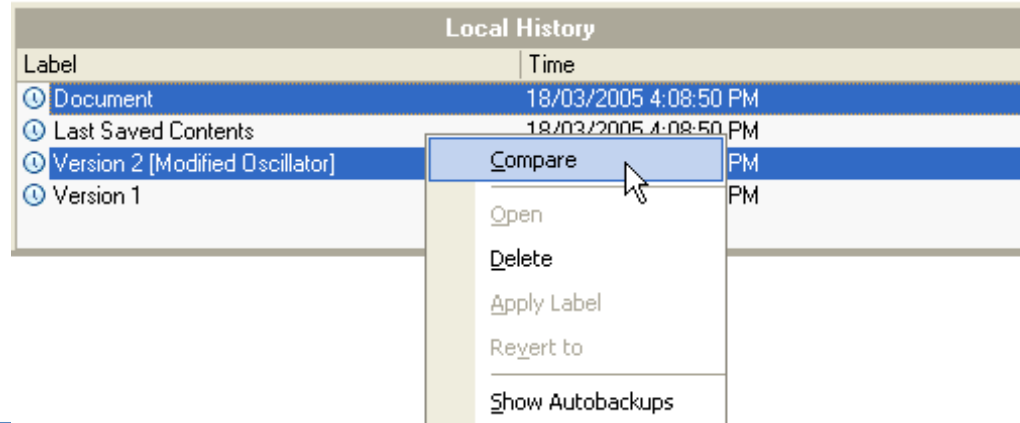
Preferences: PCB Editor

- **General:** from this dialog, we can manage some options for all PCB documents



Documents: History

- Altium's **internal history management** system allows you to maintain history and track document changes without using an external VCS
 - Allows to view differences - both physical and logical - as well as revert to previous versions
 - Works in harmony with an installed Version Control System. Individual designers can manage their own changes using the local history, with the VCS providing a complete team-oriented document management system
- It takes a copy of a file each time you perform a save, keeping all the copies in a project History folder
- Accessed through the **Storage Manager Panel**



Projects

- **The starting point for every design is a project**
 - A project is a set of design documents whose output defines a single implementation
 - Examples:
 - A PCB project includes the schematics and layout files required to manufacture a single printed circuit board
 - A FPGA project comprises the schematics and HDL files that defines the bitstream of a single FPGA
- **The project file stores:**
 - All project-relevant settings
 - A link to each document in the project via a relative reference for files on the same logical drive, or an absolute reference for files on a different logical drive
 - Links to outputs generated from the project

Types of Projects

- PCB Project (*.PrjPcb)
 - Schematics and PCB layout files, the final output are the gerbers
- FPGA Project (*.PrjFpg)
 - Schematics and/or HDL, the output is a bitstream
- Embedded Project (*.PrjEmb)
 - C and/or assembly, the output is an executable
- Core Project (*.PrjCor)
 - Schematics and/or HDL, the output is the netlist of a FPGA core
- Integrated Library (*.LibPkg) & (*.IntLib)
 - Schematic symbols plus references models: PCB footprints, circuit simulation, signal integrity and 3D mechanical models.
- Script Project (*.PrjScr): Scripts for design automation

Design Workspace

- A Design Workspace is a set of projects
 - **File ➔ Open Design Workspace** opens a Workspace and all projects it contains
 - But only one project is active at a given moment
- Useful when working with multiple projects that are closely linked in some way
 - Example: A PCB design that contains one or more FPGAs with embedded processors needs PCB, FPGA and Embedded projects
 - All projects in a Workspace can be compiled at once
- The Projects panel always displays a Workspace
 - Either the default one, or the one you have created or opened

Project: Operations

The image shows a screenshot of the Altium Designer software interface. The main window displays a project named 'MiniPC.PrjPcb' under 'Workspace1.DsnWrk'. The 'Source Documents' list includes files like 'MiniPC.PcbDoc', 'MiniPC_Func.SchDoc', and various component-specific schematic documents (e.g., 'DDR4.SchDoc', 'Ethernet1.SchDoc', 'miniPCIE_Connector.SchDoc', 'PCIE_Connector.SchDoc', 'PLL.SchDoc', 'PLL1.SchDoc'). A context menu is open over the project, listing various operations. Several callouts in blue cloud shapes point to specific items in the menu:

- Compile**: Points to the 'Compile PCB Project MiniPC.PrjPcb' menu item.
- File and project management**: Points to the 'Add New to Project', 'Add Existing to Project...', 'Save Project', 'Save Project As...', 'Close Project Documents', 'Close Project', and 'Explore' items.
- Document management**: Points to the 'Show Differences...' item.
- Differences and variants**: Points to the 'Variants...' item.
- Version control and storage management**: Points to the 'Version Control', 'Local History', 'Project Packager...', 'Project Releaser...', and 'Project Options...' items.
- Project Packager !!**: A red text callout pointing to the 'Project Packager...' item.

Project packager

Project Packager

Select Projects to Include

Select projects to be included in the project package

Select the package target

- ☒ Package focused project (MiniPC.PrjPcb)
- ☐ Package focused project tree, starting from the focused project (MiniPC.PrjPcb)
- ☐ Package workspace



Project Packager

Zip File Options

Set packaging options. This affects the files to be packaged and the path they appear in the zip file.

Zip File Name

C:\Users\Public\Documents\Altium\AD18\Examples\Mini PC\Mini PC - Motherboard\MiniPC (24-09-2018 15-44-11).zip

Directories in Zip File

- ☐ Use relative path to file drive
- ☒ Use relative paths to common parent directory of all files packaged

Generated Files

- ☐ Include (only if on the same drive as the owner project)
- ☒ Do not include

Files on Different Drive

- ☐ Include a plain copy of the file in a project sub-directory (The project is NOT modified to reference the copied files)
- ☒ Do not include

Additional Items to Include

- ☐ Output folder and all sub-folders (same drive as the project)
- ☐ All files in project directories (excluding output and history folders)
- ☐ History files (only if not using global repository)
- ☐ EDIF Files

Cancel Back Next Finish

Project Packager

Select Files to Include

Files "checked" will be packaged. The path that files appear in the package depends on the options set in the previous step. Clicking "Next" will start the package process and create the output file.

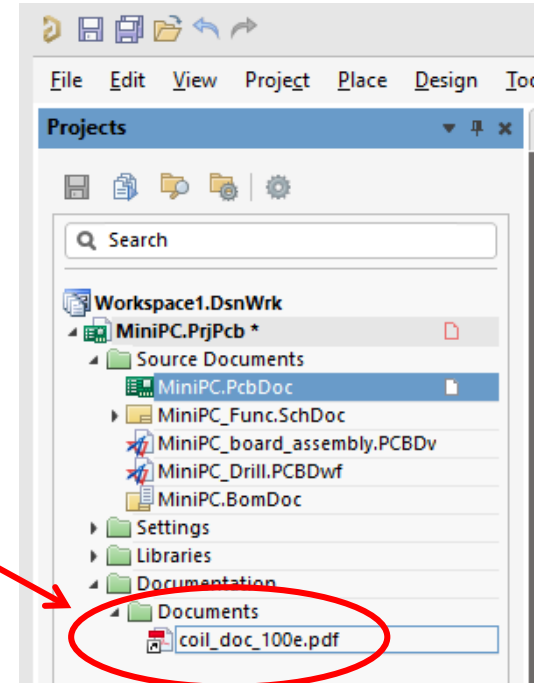
Package	File Name	Path in Zip File
☒	MiniPC.PcbDoc	MiniPC.PcbDoc
☒	MiniPC.PrjPcbStructure	MiniPC.PrjPcbStructure
☒	12V_3V3_1V8_Switches.SchDoc	12V_3V3_1V8_Switches.SchDoc
☒	12V_POWER.SchDoc	12V_POWER.SchDoc
☒	12V_to_0V9.SchDoc	12V_to_0V9.SchDoc
☒	12V_to_3V3.SchDoc	12V_to_3V3.SchDoc
☒	12V_to_3V.SchDoc	12V_to_3V.SchDoc
☒	2.5V_and_1.8V_Switches.SchDoc	2.5V_and_1.8V_Switches.SchDoc
☒	2XSFP_Arria10.SchDoc	2XSFP_Arria10.SchDoc
☒	2XSFP_Arria10.Harness	2XSFP_Arria10.Harness
☒	3.3V_and_1.8V_Load.SchDoc	3.3V_and_1.8V_Load.SchDoc
☒	3.3V_to_0.9V.SchDoc	3.3V_to_0.9V.SchDoc
☒	3.3V_to_1.0V.SchDoc	3.3V_to_1.0V.SchDoc
☒	3.3V_to_DDR4_VDD.SchDoc	3.3V_to_DDR4_VDD.SchDoc
☒	3.3V_to_VAD1.SchDoc	3.3V_to_VAD1.SchDoc
☒	3V3_to_0V9.SchDoc	3V3_to_0V9.SchDoc
☒	3V3_to_1V8.SchDoc	3V3_to_1V8.SchDoc
☒	3V3_to_2V5.SchDoc	3V3_to_2V5.SchDoc
☒	Arria10_Ground.SchDoc	Arria10_Ground.SchDoc
☒	Arria10_Power.SchDoc	Arria10_Power.SchDoc
☒	Core_Power_Decoupling.SchDoc	Core_Power_Decoupling.SchDoc
☒	DDR4.SchDoc	DDR4.SchDoc
☒	DDR4.Harness	DDR4.Harness
☒	DDR4_Arria10.SchDoc	DDR4_Arria10.SchDoc
☒	DDR4_Arria10.Harness	DDR4_Arria10.Harness
☒	DDR4_COREMMA_Arria10.SchDoc	DDR4_COREMMA_Arria10.SchDoc

Cancel Back Next Finish



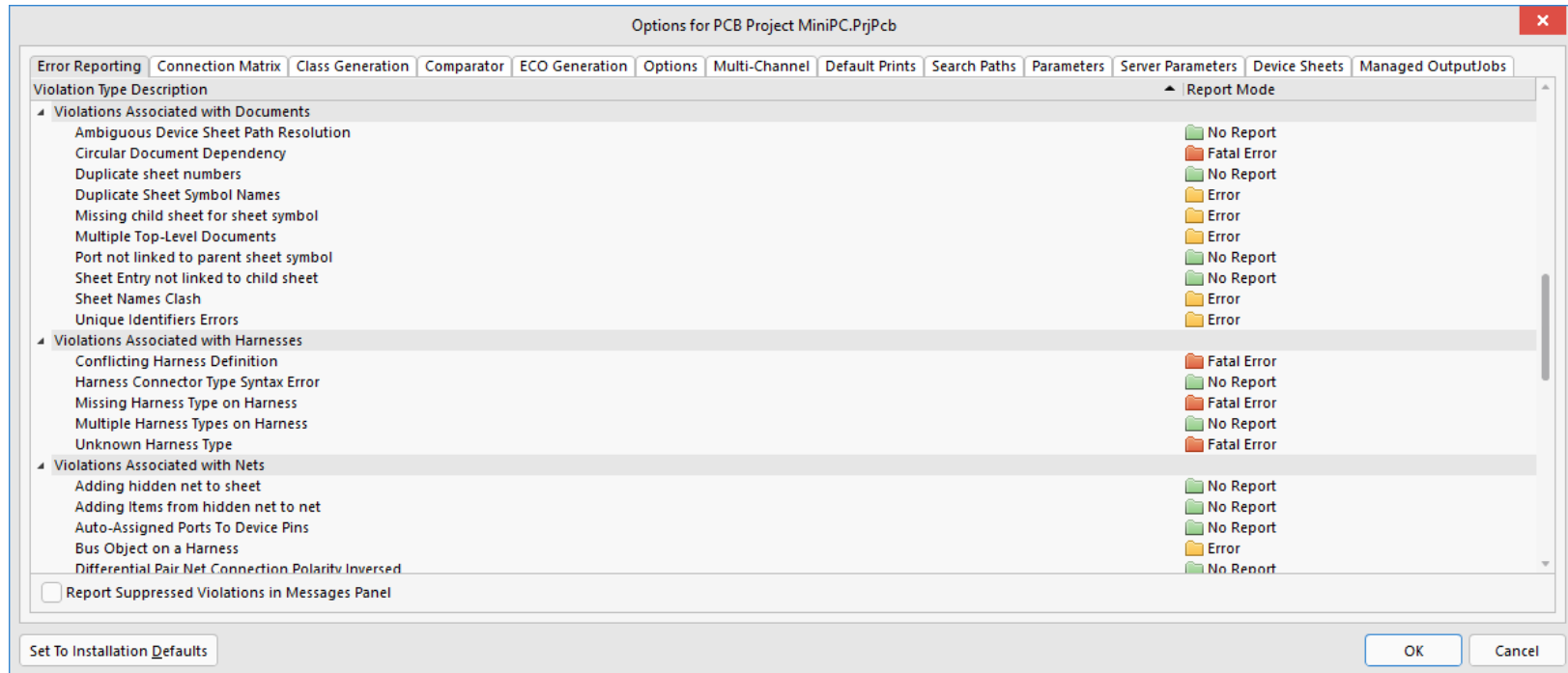
Project Documents

- Documents are automatically organized in folders:
 - Source documents, Libraries, Generated, etc.
- It is possible to add user documents to a project
 - PDFs, Word, HTML, etc.
- To change document order, simply drag and drop
 - In case of hierarchical designs, order is updated after compilation
- Documents can be shared among projects, only links are stored in projects



Project Options (1)

- Project-Specific options, as opposed to global preferences

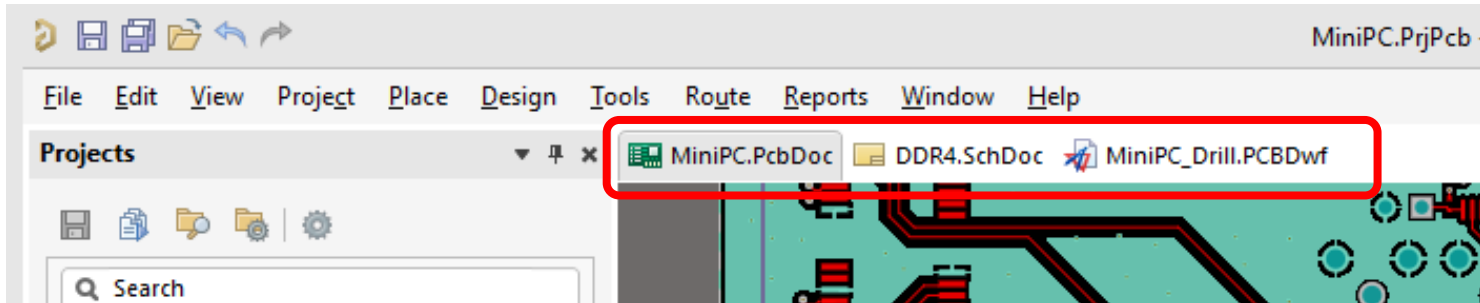


Project Options (2)

- **Error reporting:** Severity (error, warning, fatal error) of different fault conditions
- **Connection Matrix:** What is consider error when to pins are connected together
- **Class Generation:** Components and nets are organized in classes
- **Comparator:** Configuration of Find Differences
- **ECO Generator:** Select if a given difference triggers an ECO (Engineering Change Order) or not
- **Options:** Various generic options
- **Multichannel:** Configuration of multichannel designs
- **Default prints:** Output configuration for different documents being generated
- **Search paths:** For libraries
- **Parameters:** Generic parameters for the project
- **Device Sheets:** Configuration of projects using device sheets

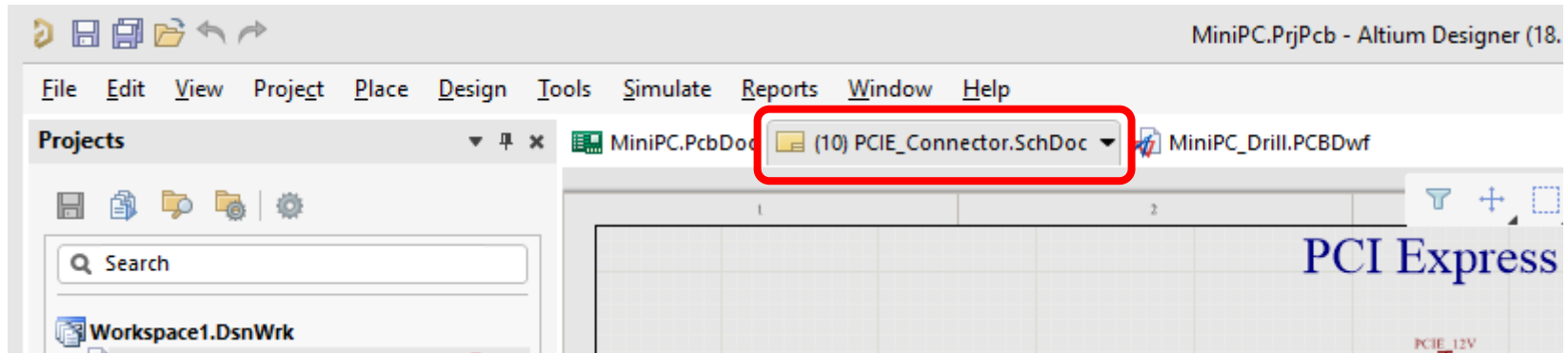
Working with Documents

- Multiple documents may be opened simultaneously.
 - Each open document has its own tab at the top of the design window on the Document Bar
 - Only one document may be active within the main design window
- Click at a document's tab to show it on the main design window
- **CTRL + TAB** and **CTRL + SHIFT + TAB** cycle forward or backward through all open documents

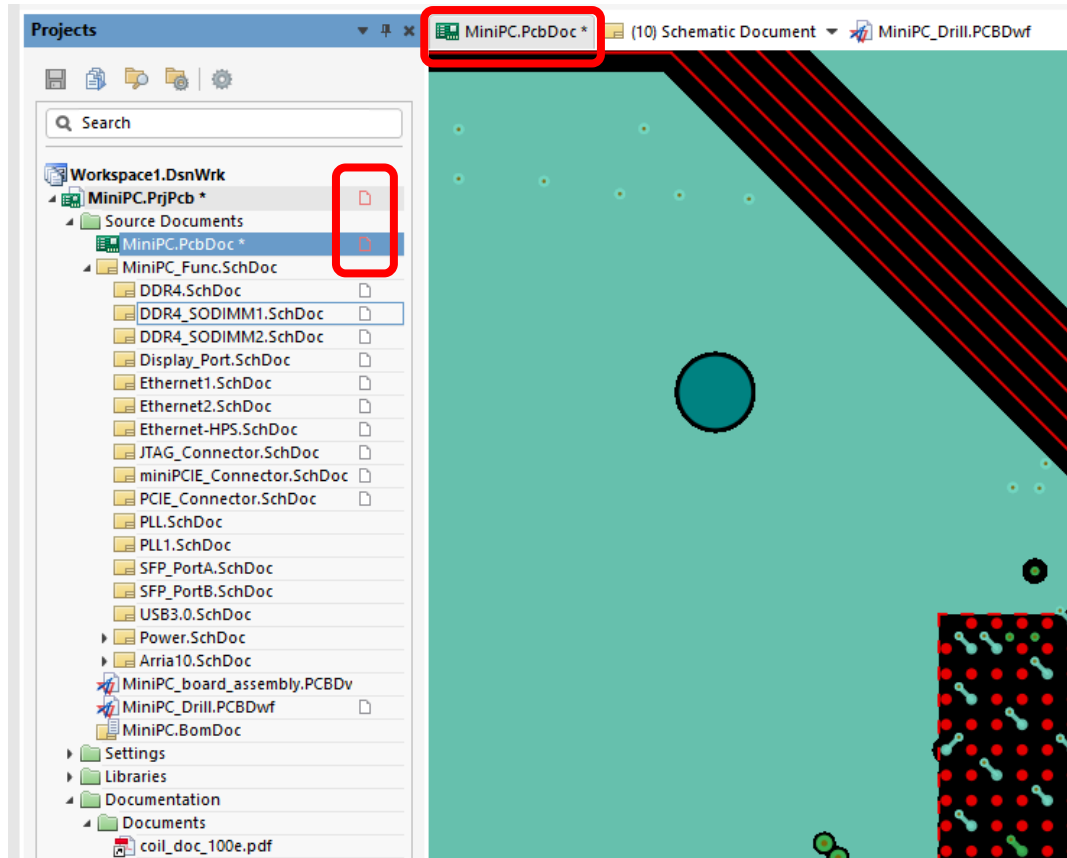


Working with Documents: Groups

- When a large number of documents are open, they are grouped under one tab
 - Grouping is enabled/disabled in the Document Bar options of the **System** ➔ **View** page of the Preferences dialog
 - Documents can be grouped by project or by kind



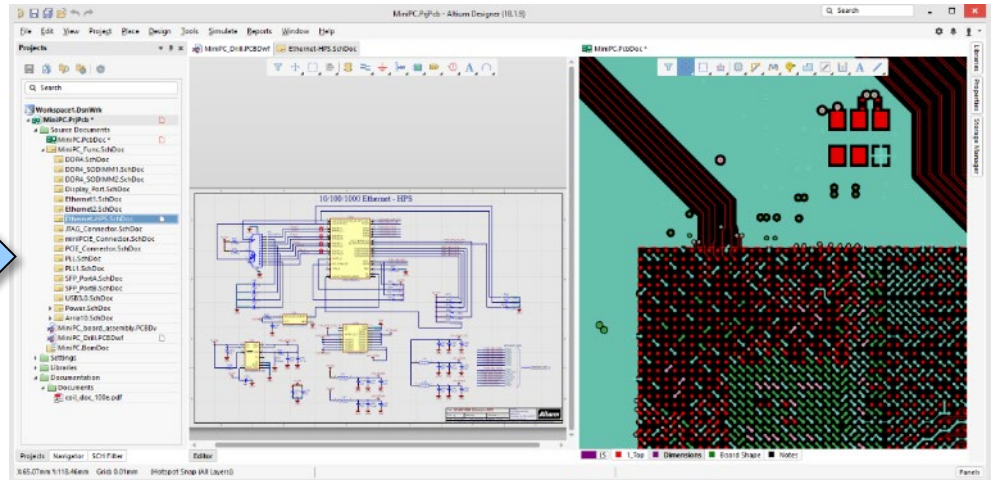
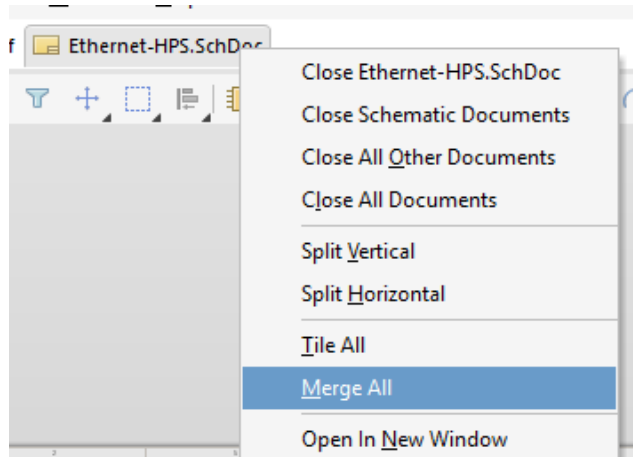
Document Status



- By default, icons show the status of documents in the Project panel
- Icons in red are modified but not saved yet documents
- Modified but not saved yet documents have a mark (*) in the documents tab

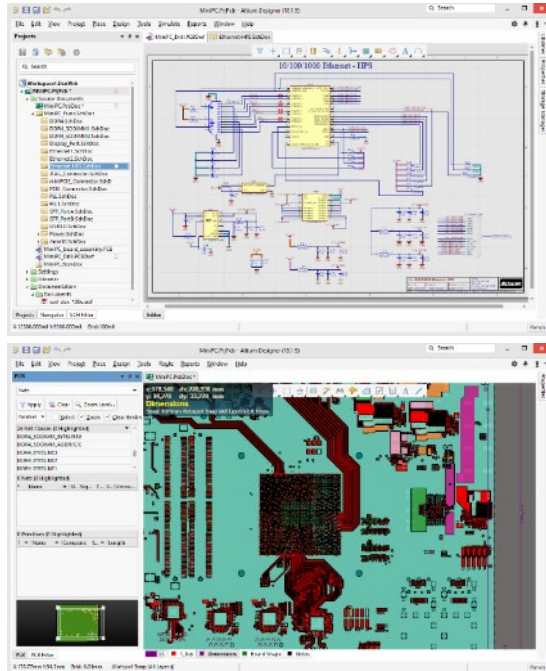
Multiple Design Windows

- Right-click on a document tab to split the main design window
 - You can split it horizontally or vertically
 - Drag and drop to move documents from one window to the other
- Choose **Merge all** to un-split the window



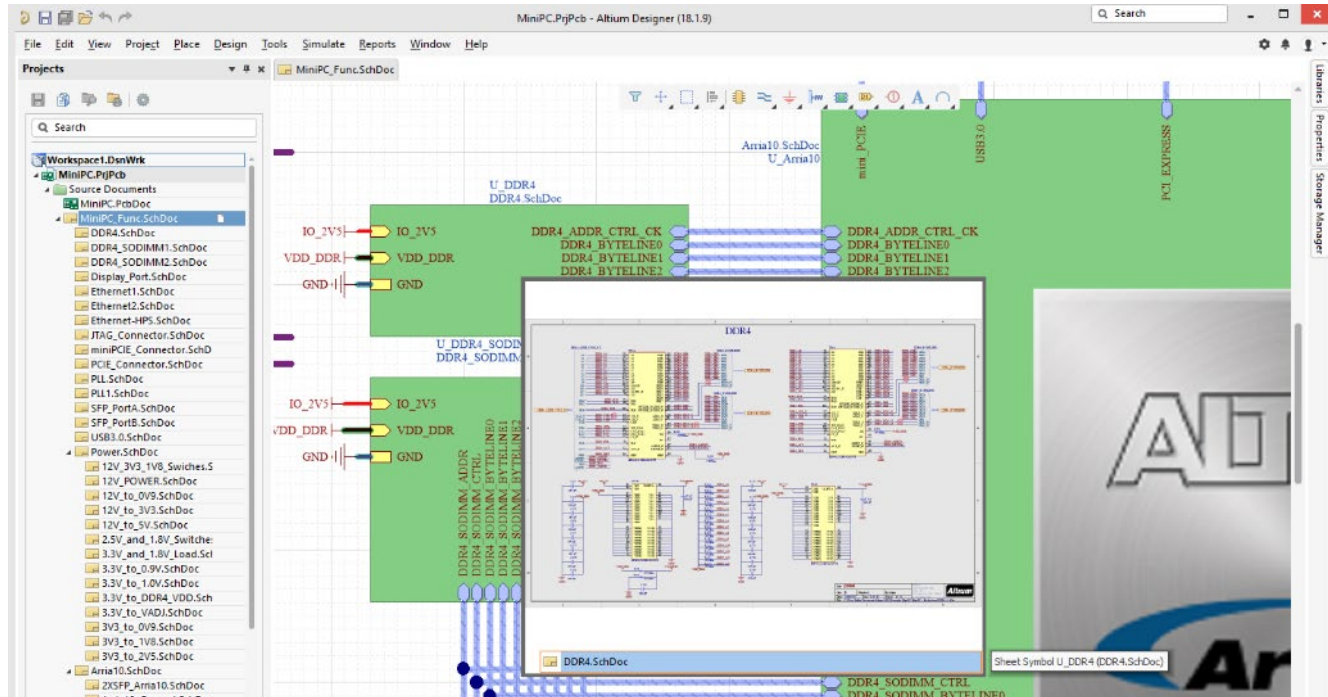
Multiple Altium Windows

- Right click on a document tab and select **Open in New Window**
- Perfect for working with multiple monitors



Design Insight

- Allows you to preview documents, sheets, connections...
 - Preferences ➔ System ➔ Design Insight to define the behavior





Agenda

- Altium Designer Environment
- **Drawing Schematics**
- PCB Editor

Schematic Options

Accessible in the Properties panel, when no other object is selected

Units

Visible and snap grids

Sheet color

Sheet size & template selection

More formatting

The screenshot shows the 'General' section of the Schematic Properties panel. It includes settings for Units (mm and mils), Visible Grid (100mil), Snap Grid (100mil), Snap to Electrical Object (checked), Snap Distance (100mil), Document Font (Times New Roman, 10), Sheet Border (checked), and Sheet Color (white). The 'Page Options' section includes 'Formatting and Size' with Template, Standard, and Custom options, and 'Margin and Zones' with Show Zones (checked), Vertical (4), Horizontal (8), and Origin (Upper Left). At the bottom, there are tabs for Libraries, Properties, and Panels.

General

Units

☐ mm ☒ mils

Visible Grid

☐ Snap Grid G

☒ Snap to Electrical Object

Snap Distance

Document Font

Sheet Border ☒

☐ Sheet Color

Page Options

Formatting and Size

☒ Template

Width: 32000mil | Height: 2000

Margin and Zones

☒ Show Zones

Vertical

Horizontal

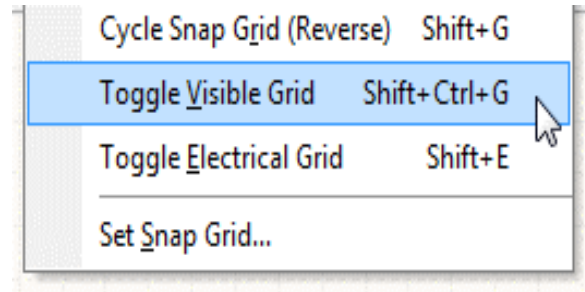
Origin

Nothing selected

Grids

- **There are three types of grids in Altium:**
 - **Snap grid:** Grid used when placing objects
 - **Visible grid:** Visible as lines or dots when zoom is big enough
 - **Electrical grid:** Enables connections on off-grid objects
- When moving an electrical object, if it falls within the electrical grid range of another object that you could connect to, it will snap to the fixed object and a hotspot (red cross) will appear
- Snap grid is always active, others are enabled in the grid dialog

View -> Grids commands



Placing Objects

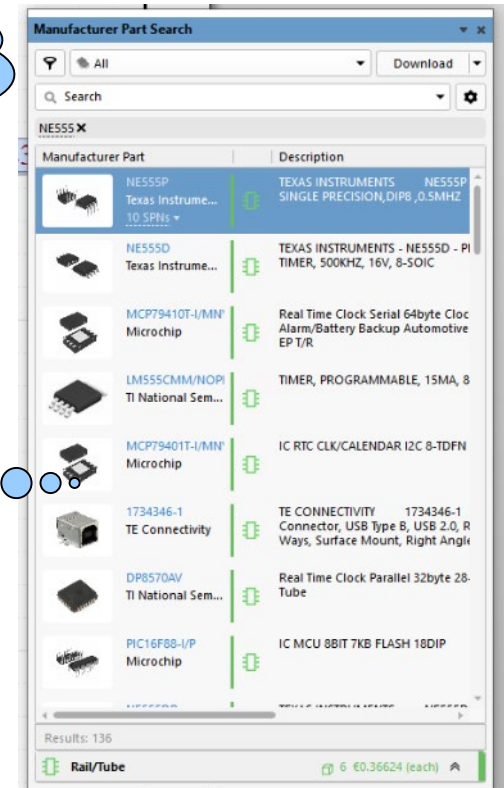


From the
floating toolbar

From the
Place menu

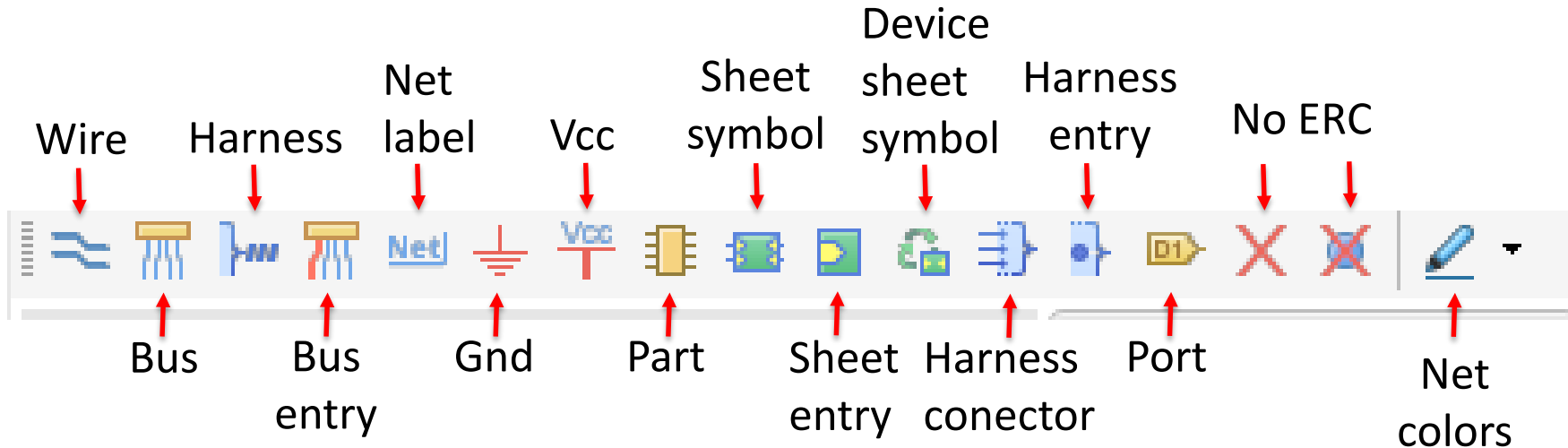
From the
Manufacturer
Part Search Panel

Also from the
wiring toolbar!



Electrical Objects

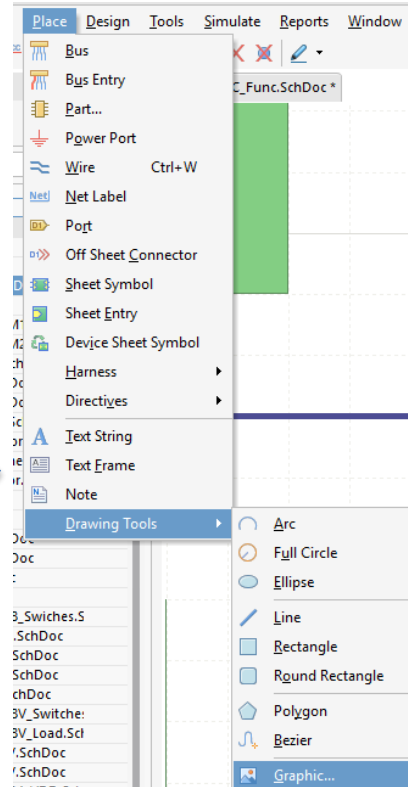
- Objects that define the functionality of the circuit
 - Parts, hierarchies and connections (**Wiring** toolbar)



Graphical Objects

- Objects used for documentation and drawing

Text String
Text frame
Note

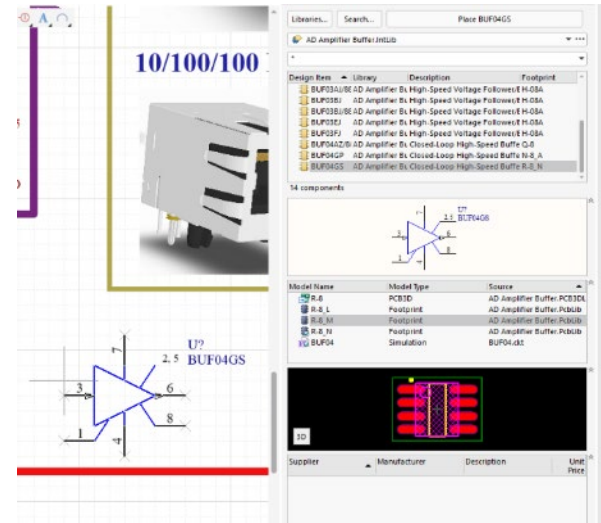


- Arc
- Full circle
- Ellipse
- Line Rectangle
- Round rectangle
- Polygon
- Bezier
- Graphic



Placing Objects: Tricks

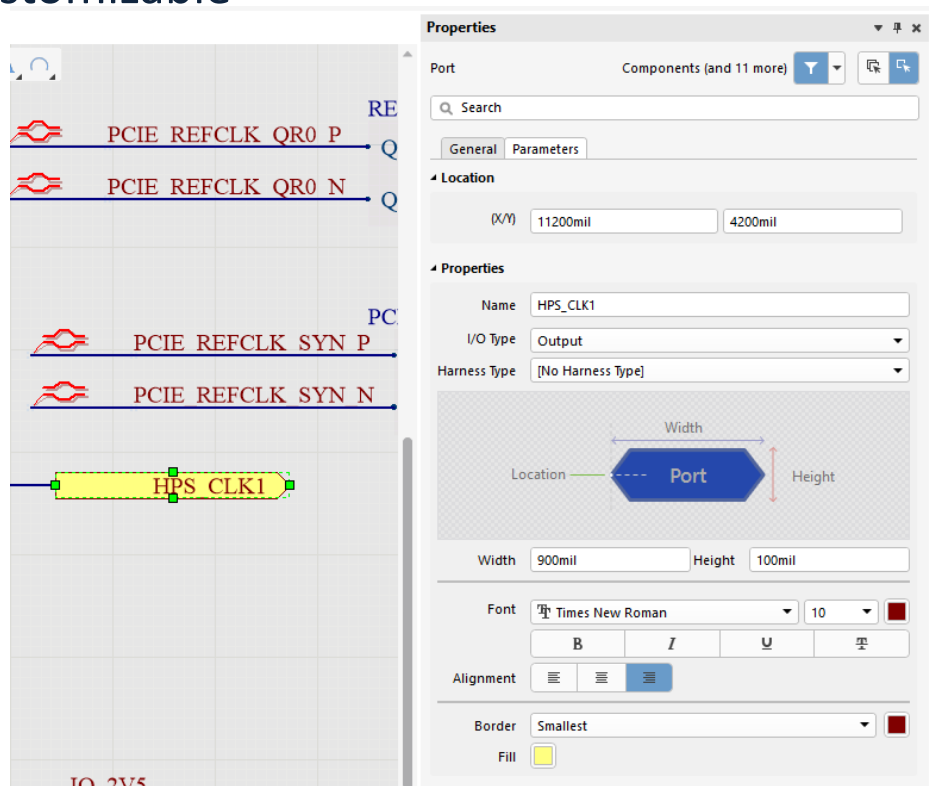
- When placing an object:
 - SPACE rotates it
 - X flips horizontally
 - Y flips vertically
- Zooming
 - Ctrl + Mouse Wheel or (Wheel button + Mouse up and down)
- Panning
 - Mouse Wheel pans horizontally, Shift + Wheel pans vertically



Placing Objects: Properties

- All drawing objects are totally customizable

– Colors, line thicknesses, aspect, electrical properties, etc.



Move and Drag

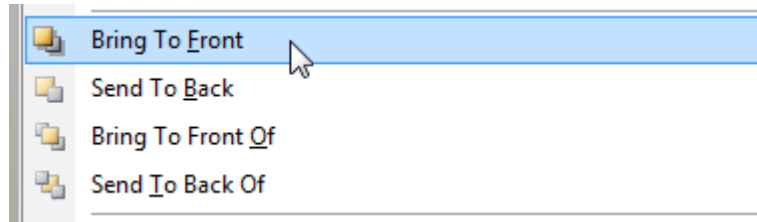
- Moving or dragging?
 - Move does not keep connections, but dragging does
 - When an object is selected, click-and-drag the mouse over it moves the object, not drags it: connections break
- “Schematic Preferences ➔ Graphical Editing ➔ Always Drag” selects default action
- While objects are being moved:
 - SPACEBAR to rotate the object anti-clockwise
 - SHIFT+SPACEBAR to rotate the object clockwise
 - X or Y keys to flip the object along the X-axis or Y-axis
 - Hold down ALT to constrain movement to the horizontal or vertical axis.
- Moving selected objects
 - CTRL + Arrow keys in small increments (snap grid)
 - CTRL + SHIFT + Arrow keys in big increments (10x snap grid)
- **Edit ➔ Move ➔ Drag** moves any single object while **Edit ➔ Move ➔ Drag Selection** moves a set of objects previously selected
 - Press the SPACEBAR or SHIFT+SPACEBAR while dragging to change the wiring mode
 - CTRL + SPACEBAR to rotate
 - Other keys as in moving (SPACEBAR, X, Y, ALT)

Common Concepts

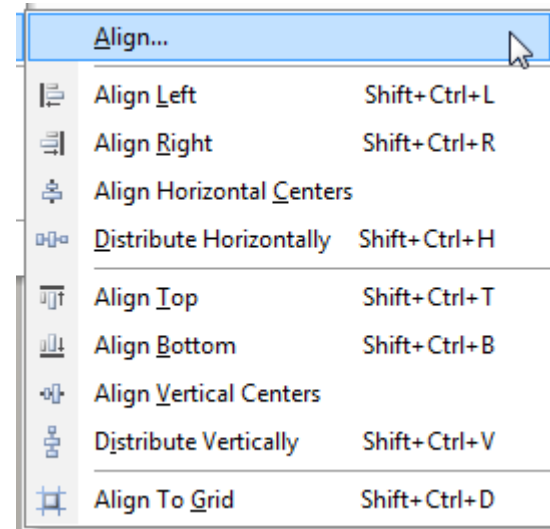
Same concepts as in design programs are used:

- All objects are drawn in a layer, you can push layers front or back in order to show objects or hide them under other objects

Edit -> Move commands

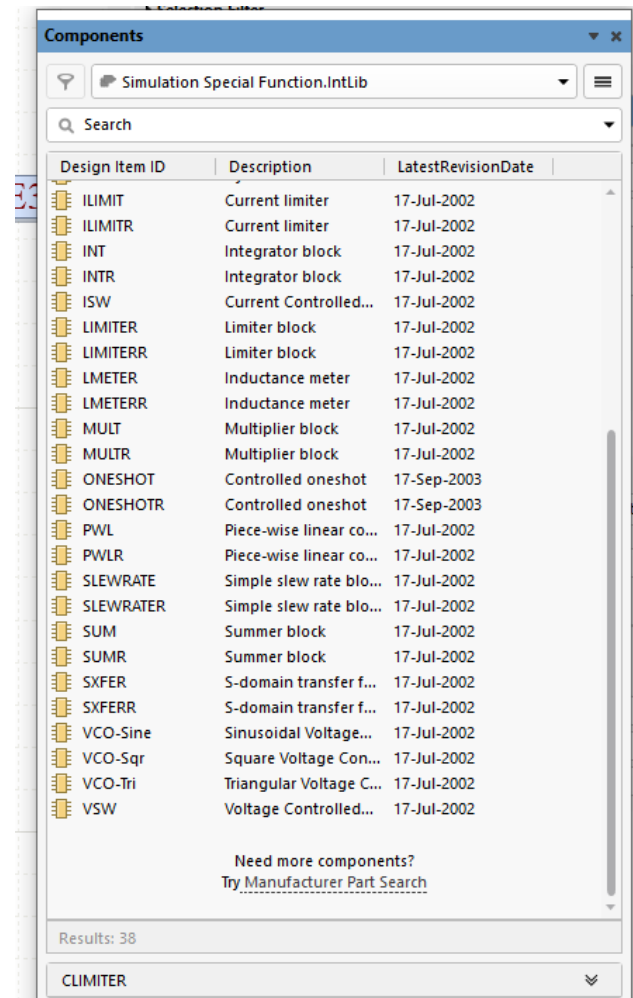


Edit -> Align commands



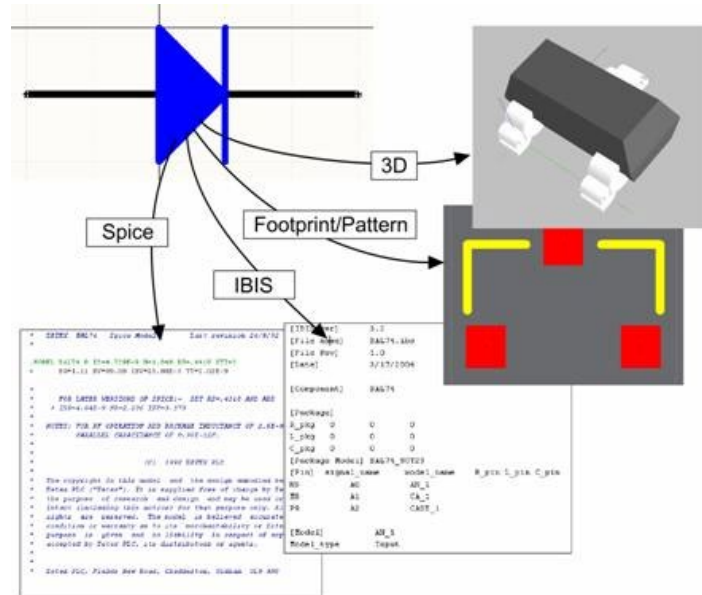
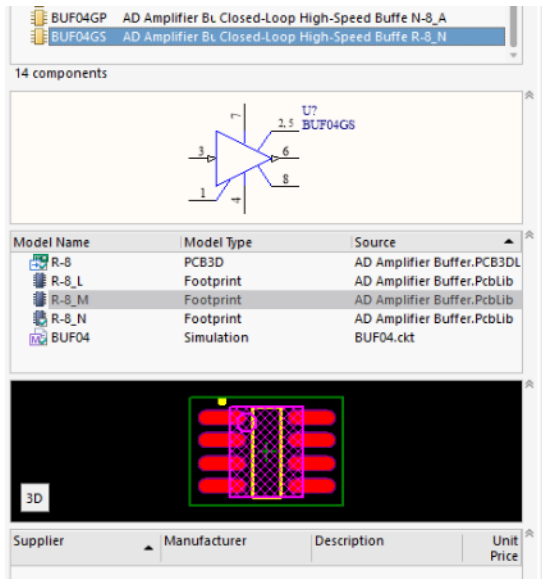
Placing Parts

- The basic command is **Place ➔ Part** or 
- Automatically opens the **Components** panel
- Libraries can be added and installed
- Steps to place a part:
 - Choose the component from the libraries
 - Make sure that the footprint is the right one
 - For multi-part packages, choose the part
 - Comment = Part Value



Altium Libraries

- At the schematic stage, the design is a collection of components that have been connected logically
- To test or implement the design it needs to be transferred to another modeling domain, such as simulation, PCB layout, etc.

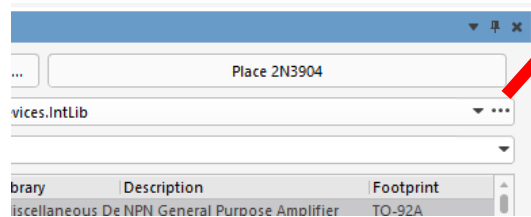
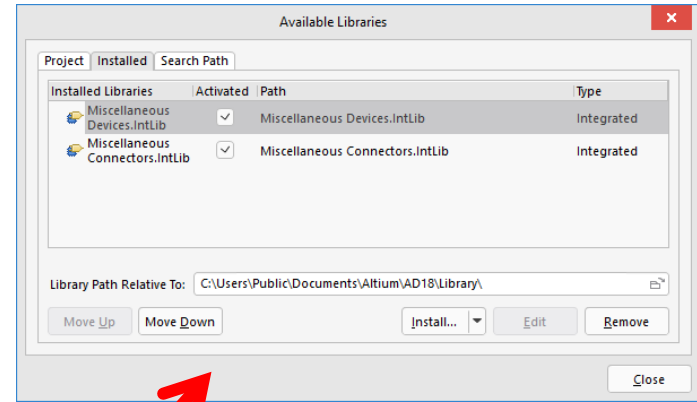


Installing Libraries

- Three sources for libraries:
 - Local libraries for the **Project**
 - Libraries **Installed** in the Altium Designer Environment
 - Search **Path** to look for libraries

- Search path is not for integrated or schematic libraries

- Just for model libraries (PCB footprints, 3D models, SPICE and VHDL models)
- And also for database and SVN libraries
 - DBLib and SVNDBLib



Component Properties

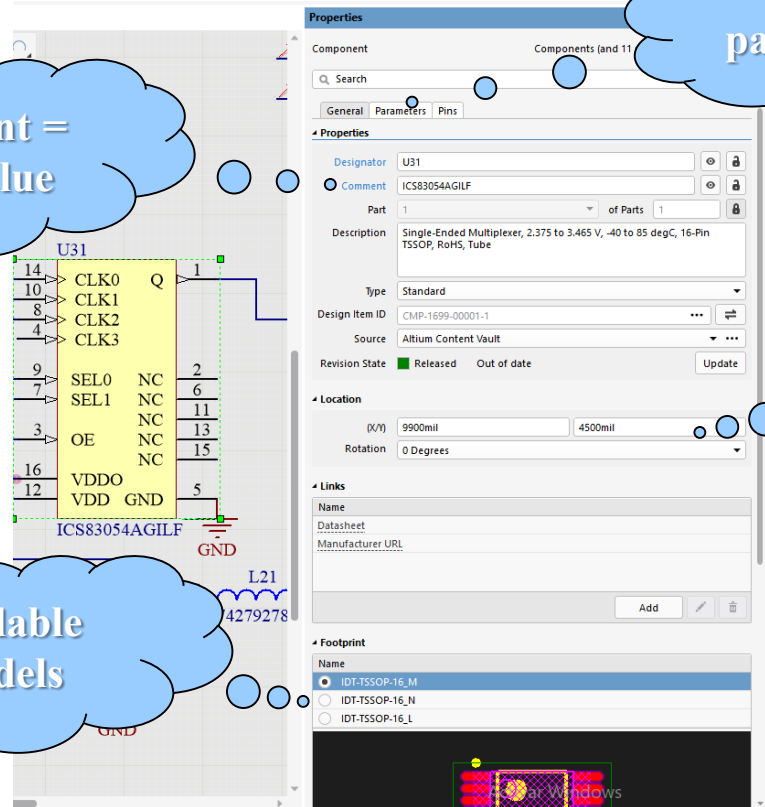
- Properties panel

Comment =
Part value

User
parameters

Position and
aspect

Available
Models

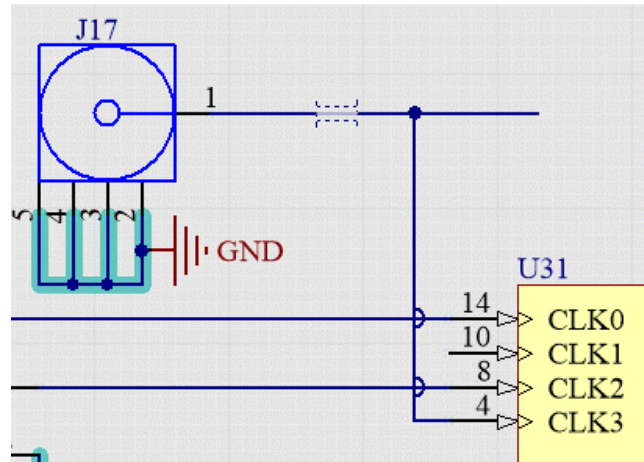
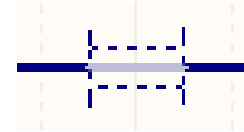
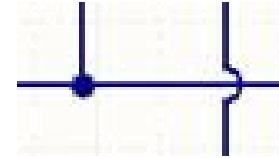


Placing Wires

- Wires  represent an electrical connection between points
 - Be careful not to use Lines by mistake
- When placing a wire, if the wire falls within the electrical grid range of another electrical object, the cursor will snap to the fixed object and a 'hot spot' (red cross) will appear.
 - Hot spot guides automatically snap the cursor to electrical connection points
 - The electrical grid should be slightly smaller than the snap grid
- 4 wiring modes, selected by SHIFT + SPACEBAR
 - 90 Degree
 - 45 Degree
 - Any Angle
 - Auto Wire
- Right-click, or key ESC, exits the wire placement mode
- BACKSPACE deletes the last vertex placed

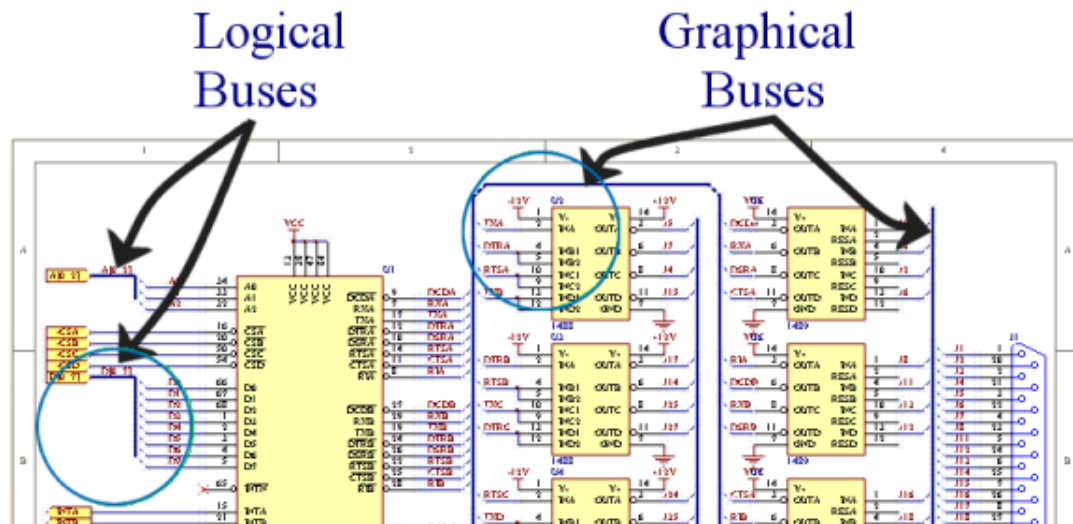
Placing Wires (2)

- Junctions and Cross-Overs
 - Junctions are automatically placed
 - Preferences-> Schematic-> General-> Display Cross-Overs allows to turn cross-overs display on/off
- **Edit ➔ Break Wires** is used to break connections
 - SPACEBAR sets the cutter size and TAB its options



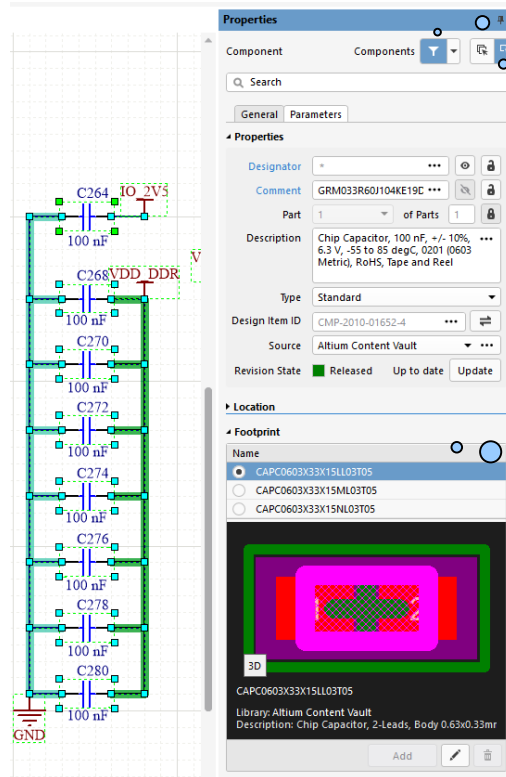
Buses

- Logical buses are used for connectivity
 - Wires have Net Labels ended by a number (e.g. sigX)
 - Bus has a net identifier of the for sig[X.Y]
- Graphical buses are used for clarifying the schematic
 - Buses do not have any net identifier



MODIFYING OBJECT PROPERTIES

- **Properties panel** combined with selection filters allows to modify one or multiple objects
- **Forget the Inspector in previous Altium versions (up to 17), replaced with the Properties panel**
- Allow powerful actions like “Change footprint in all 100 nF capacitors in this sheet to 0402 size”



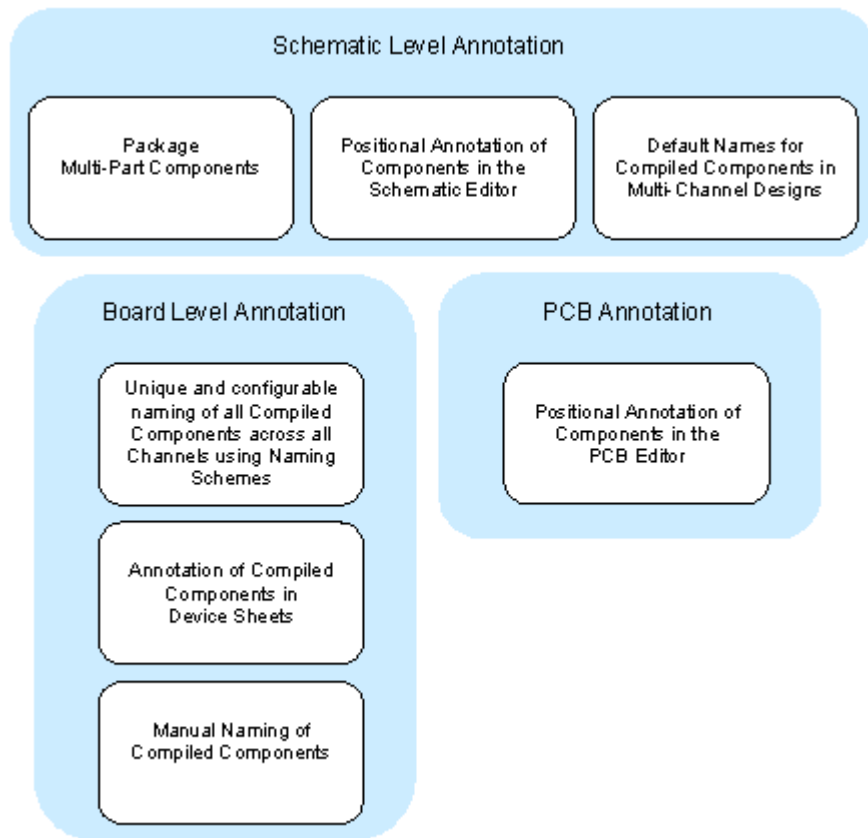
Filter

Current or open documents

Changes apply to all selected objects

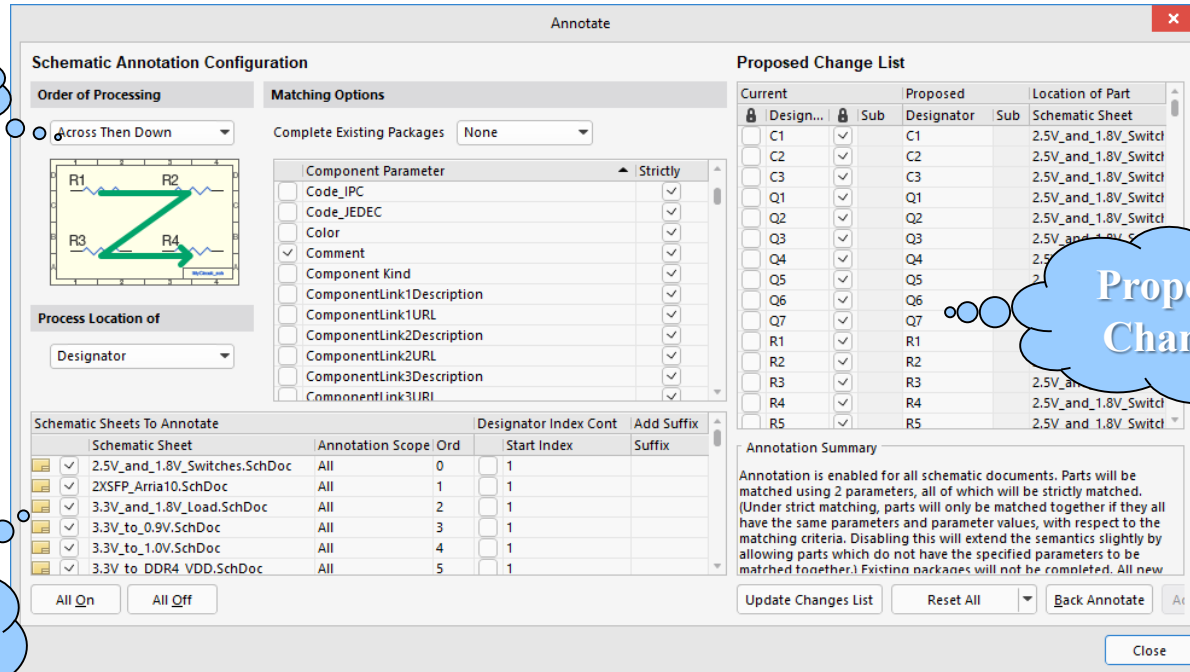
Annotation

- Three approaches:
 - Schematic Level Annotation
 - Board Level Annotation
 - PCB Annotation
- Schematic Level is always the first step
- Board Level annotates after compilation, useful for hierarchical designs
 - Compulsory if device sheets are used in the design
- PCB allows for positional PCB annotation



AnnotatIOn

- Main dialog: Tools ➔ Annotation ➔ Annotate Schematics

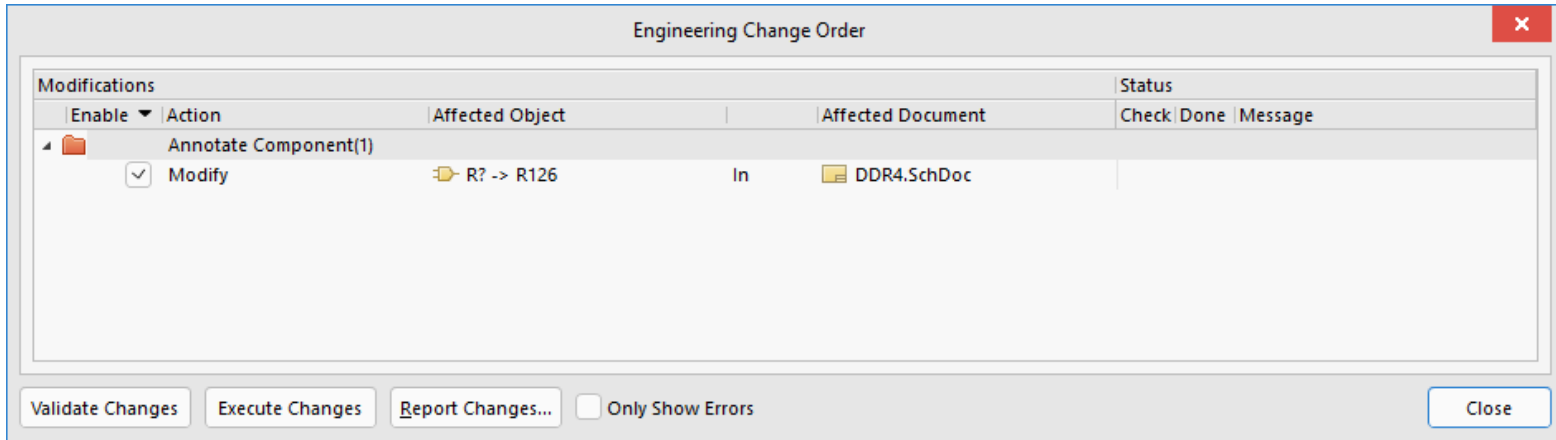


Sheets to Annotate

Proposed Changes

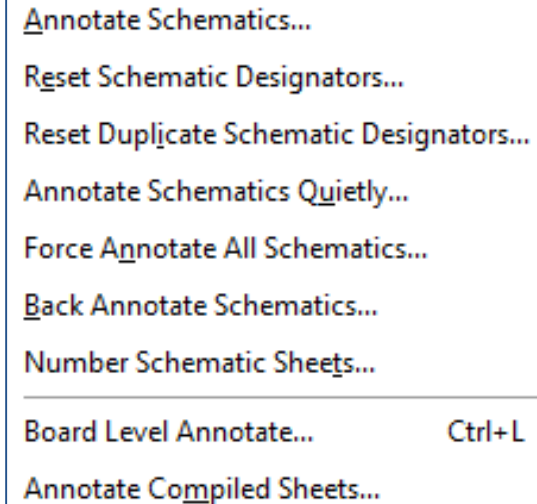
Annotate: Accept Changes

- In the previous screen, you can check the proposed annotation.
 - If you like it, press **Accept Changes (Create ECO)**
- The Engineering Change Order (ECO) dialog appears to **Validate** and **Execute Changes**



AnnotatIOn: All Tools

- **Annotate Schematics**: Main dialog
- **Reset Schematic Designators**: Resets all references to ?
- **Annotate Schematics Quietly**: Updates ? references, but not duplicated ones
- **Force Annotate All Schematics**: Solves all problems, including duplicates
- **Back Annotate Schematics**: Reads changes from the PCB Level Annotation
- **Number Schematic Sheets**: Kind of annotation for Sheets
- **Board Level Annotate**: Name says all
- **Annotate Compiled Sheets**: Number sheets after compilation

A screenshot of a software menu titled 'Annotate Schematics...'. The menu contains several options: 'Reset Schematic Designators...', 'Reset Duplicate Schematic Designators...', 'Annotate Schematics Quietly...', 'Force Annotate All Schematics...', 'Back Annotate Schematics...', 'Number Schematic Sheets...', 'Board Level Annotate...' (with a keyboard shortcut 'Ctrl+L' to its right), and 'Annotate Compiled Sheets...'. The menu is enclosed in a thin blue border.

Annotate Schematics...
Reset Schematic Designators...
Reset Duplicate Schematic Designators...
Annotate Schematics Quietly...
Force Annotate All Schematics...
Back Annotate Schematics...
Number Schematic Sheets...
Board Level Annotate... Ctrl+L
Annotate Compiled Sheets...

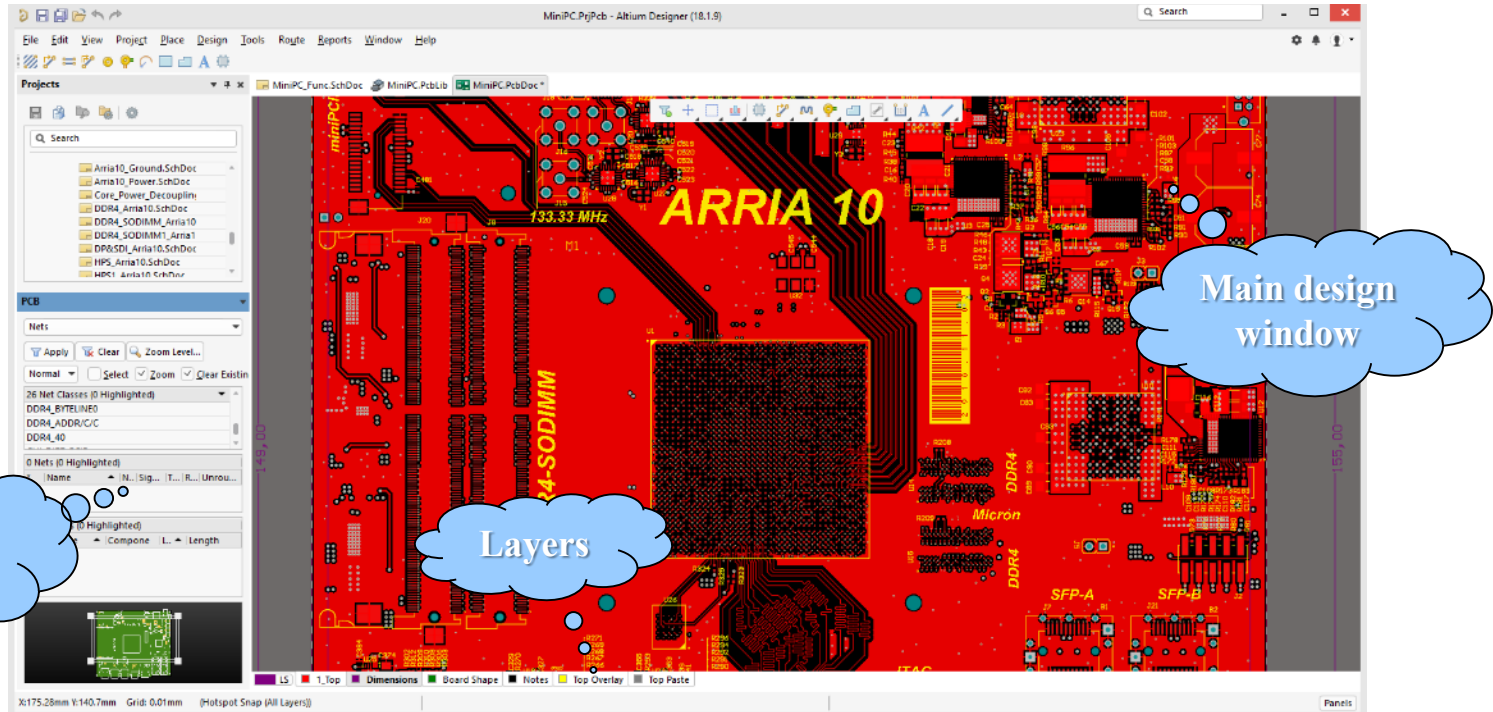


Agenda

- Altium Designer Environment
- Drawing Schematics
- **PCB Editor**

PCB Editor

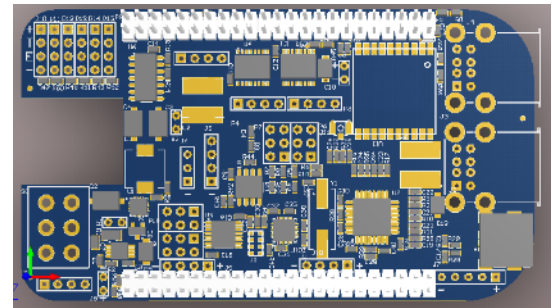
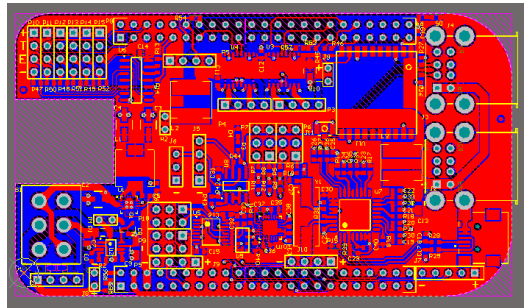
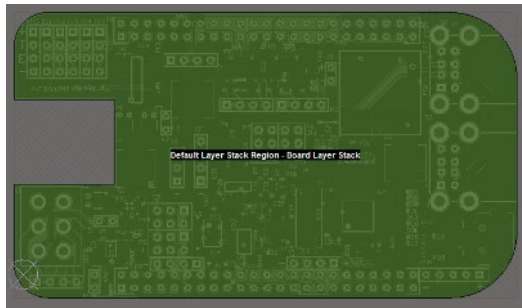
- The PCB Editor allows you to create, edit and verify the PCB design, as well as generate the output files required to manufacture the PCB



PCB Editor

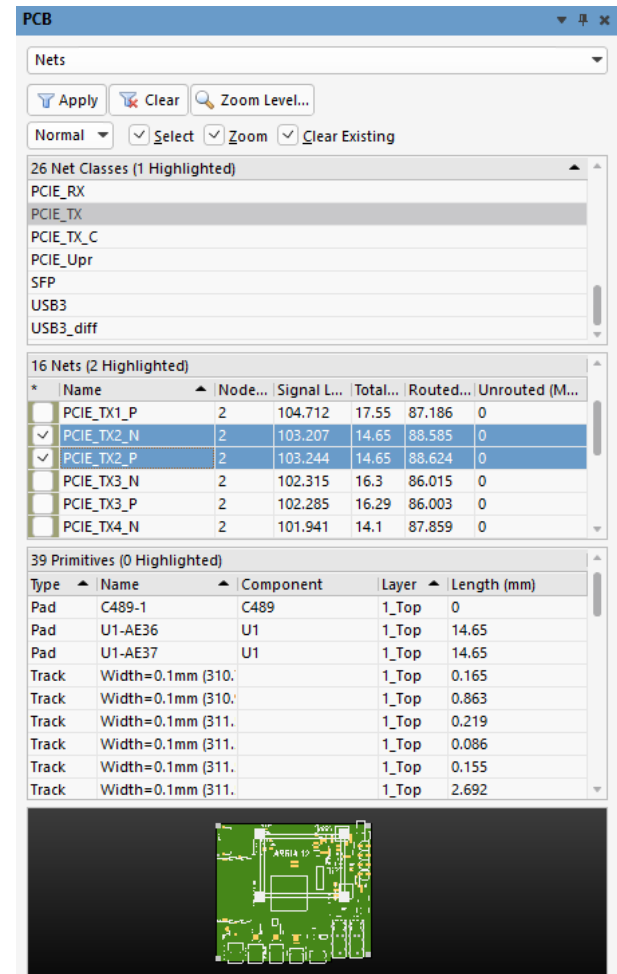
Altium has 3 different views modes for PCB editor

View	Project	Design	Tools	Auto R
	Board Planning Mode		1	
	2D Layout Mode		2	
	3D Layout Mode		3	



PCB Panel

- The PCB panel allows you to browse the current PCB design using various filter modes to determine which object types or design elements are listed, highlighted or selected
 - The top drop-down list in the panel is where you select the panel mode. You can define a browsing mode - either nets, components, design rules, 3D models (3D bodies) or polygons - or you can use the From-To, Split-Plane, Differential Pairs, and Hole Size Editors
 - The bottom window shows the whole board and the current position of the main design window



Board Options

- Properties Panel

The screenshot shows the 'Properties' panel in a PCB design application. The panel is divided into several sections:

- Board**: Components (and 12 more)
- Selection Filter**: Search bar
- Snap Options**:
 - ☒ Snap To Grids
 - ☐ Snap To Guides
- Snap to Object Hotspots**: Shift+E
 - Buttons: All Layers, Current Layer, Off
 - Snap Distance: 0.1mm
 - Near Range: 25.4mm
- Objects for snapping**:

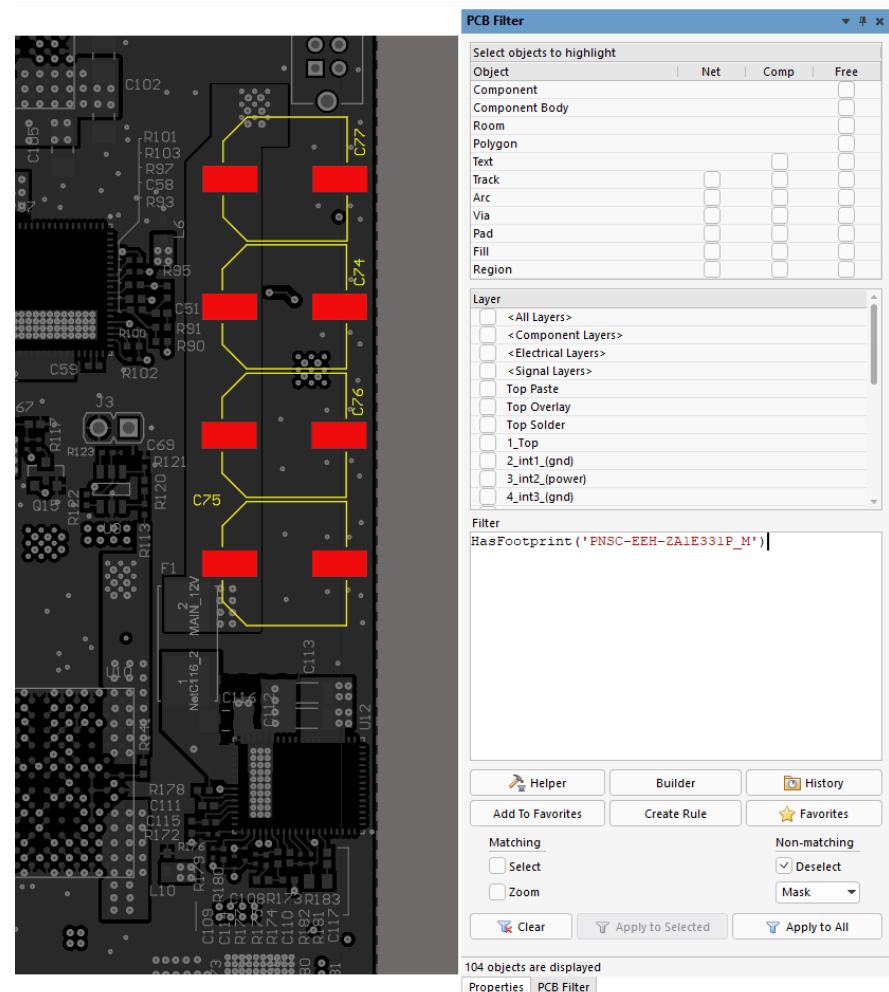
On/Off	Objects	Near	Far
☒	All Objects	☒	☒
	Pads	☒	☒
	Vias	☒	☒
☐	Arcs	☒	☐
	Tracks	☒	☐
	Regions		
	Fills		
- Board Information**:
 - Grid Manager
 - Guide Manager
- Other**:
 - Units**: mm (selected), mils
 - Polygon Naming Scheme: NET NAME_LXX_PXXX
 - Designator Display: Physical
 - Route Tool Path: Layer not defined

Callouts in the image:

- Snap**: Points to the Snap Options section.
- Board info**: Points to the Board Information section.
- Grids**: Points to the Grid Manager option.
- Units**: Points to the Units section.
- Multichannel**: Points to the Polygon Naming Scheme dropdown.

PCB Filter Panel

- The PCB Filter panel allows you to construct filters through the creation of logical queries
- A defined filter can then be applied to the active PCB document, allowing you to select, apply masking/dimming and zooming
- Selected objects can be edited (Properties panel)



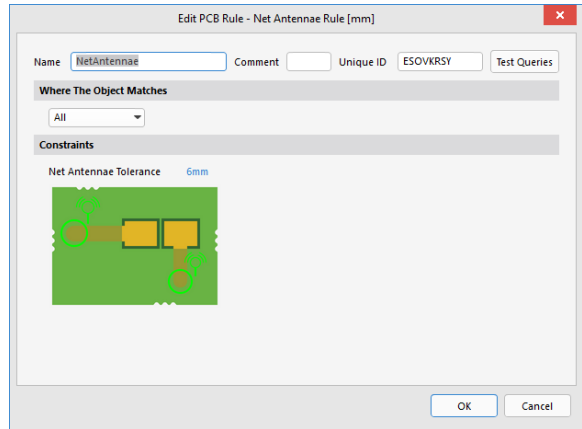
Violation

- A violation object marks an instance in the design where a particular design rule is currently being violated by one or more design objects.
 - The violation itself is highlighted in the workspace through the use of a DRC Error Marker

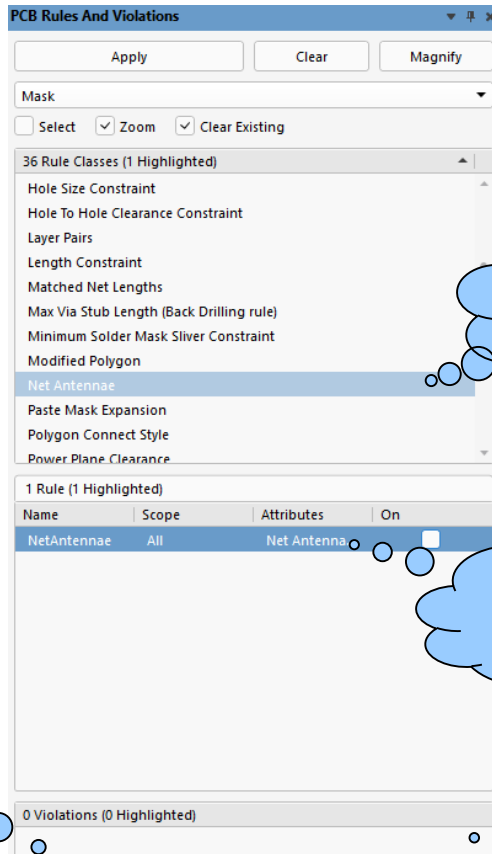


- This type of object is placed automatically by the Design Rule Checking features
 - Both Online and Batch
 - It is not a design object that can be accessed and placed by the user

PCB Rules and Violations Panel



Highlights
violations to
design rules

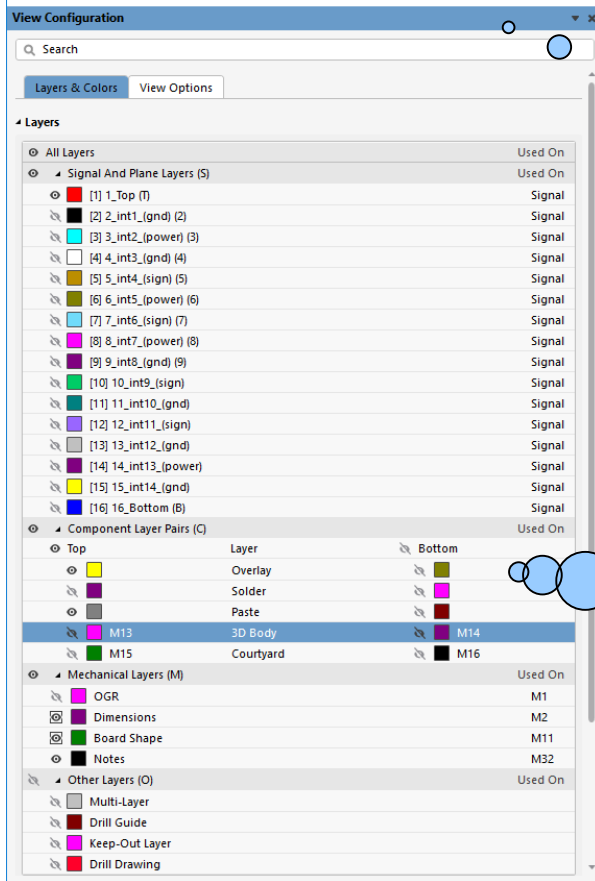


Can run DRC
for the rule
here

Shows design
rule

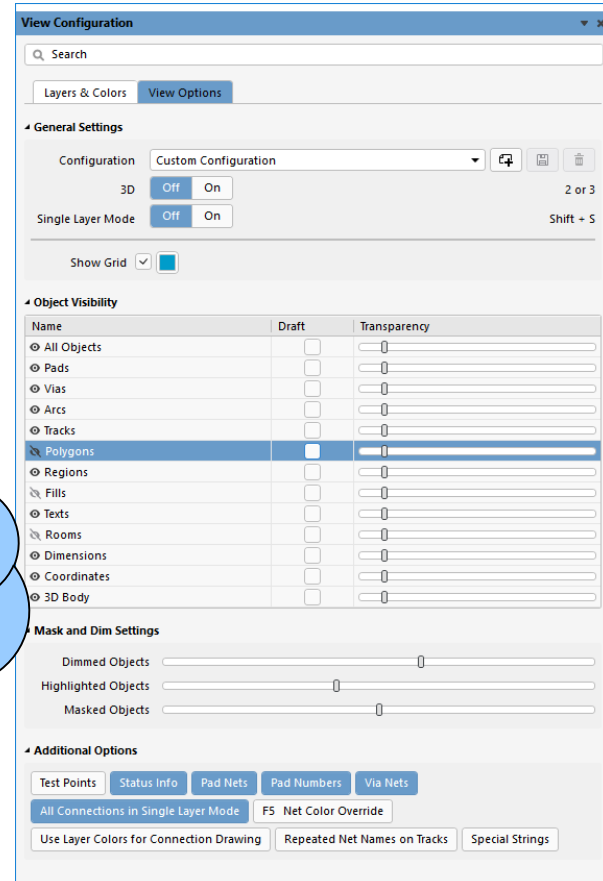
Violations can
be waived

View Configurations PANEL: 2D

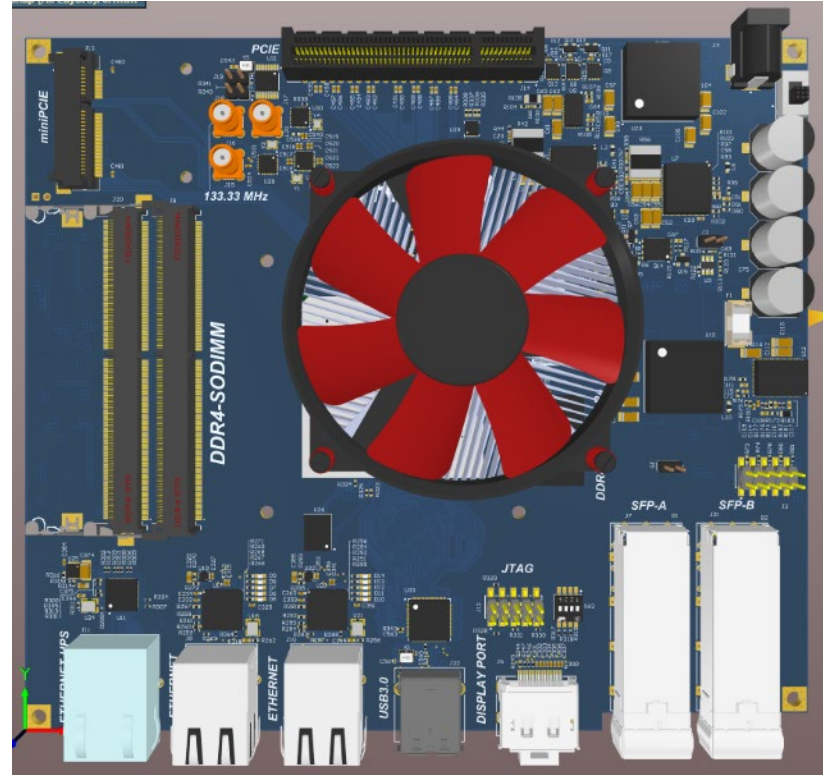
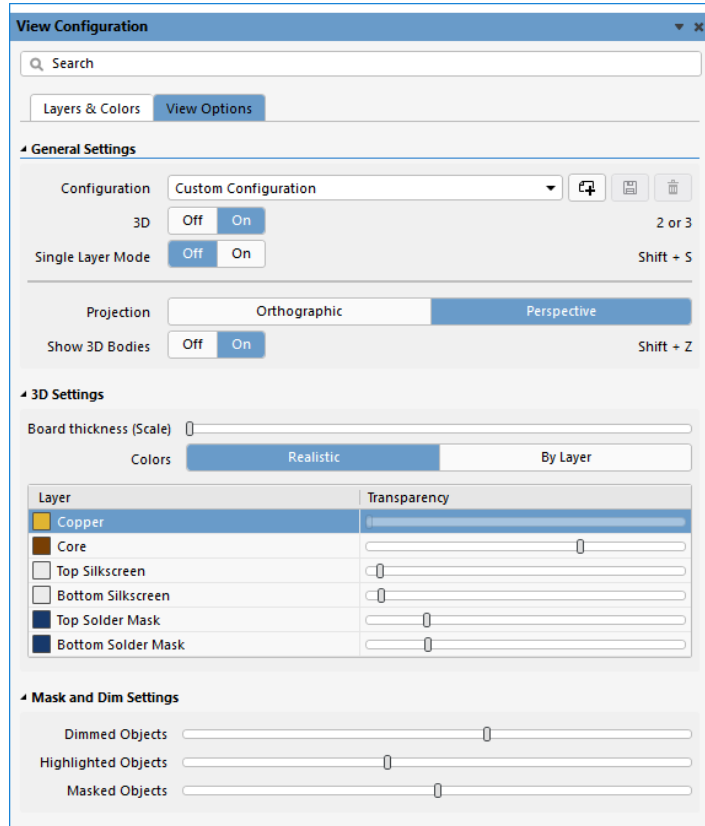


Press "L"
to open

Colors and
visibility for
all layers and
object types

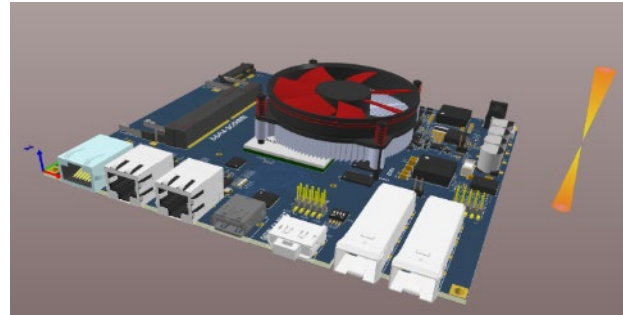
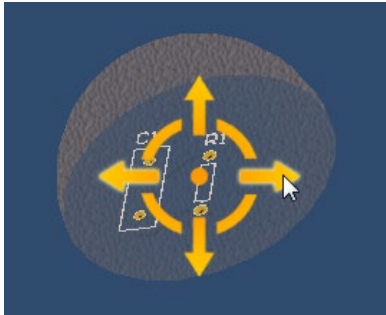


View Configurations PANEL: 3D



3D Rotation

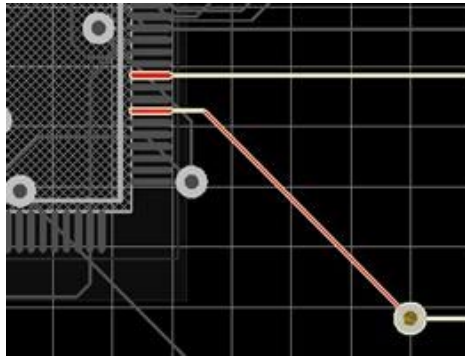
- Hold down **SHIFT** to enter 3D rotation mode
 - This is represented as a directional sphere at the cursor position:



- Right-drag sphere **Center Dot** with the mouse for full floating view - rotate in any direction
- Right-drag sphere **Horizontal Arrow** with the mouse to rotate the view about the Y-axis
- Right-drag sphere **Vertical Arrow** with the mouse to rotate the view about the X-axis
- Right-drag sphere **Circle Segment** with the mouse to rotate the view in the Y-plane

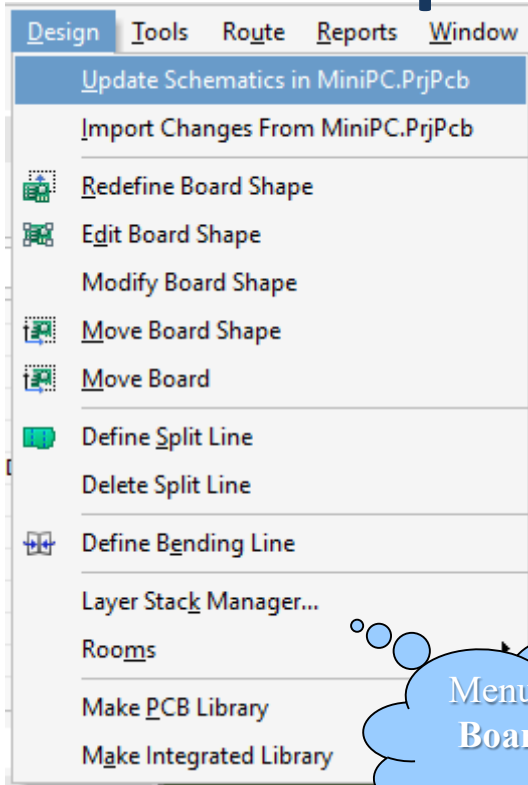
Masking

- Mask highlight can be applied in several occasions:
 - when applying a query from the PCB Filter panel,
 - when browsing design objects using the PCB panel,
 - when interactively routing



- The **Mask Level** can be adjusted from the **View** menu
- Masking and dimming is determined by the **Highlighting Options** set in the **PCB Editor - Display** page of **Preferences**

Board Shape Operations



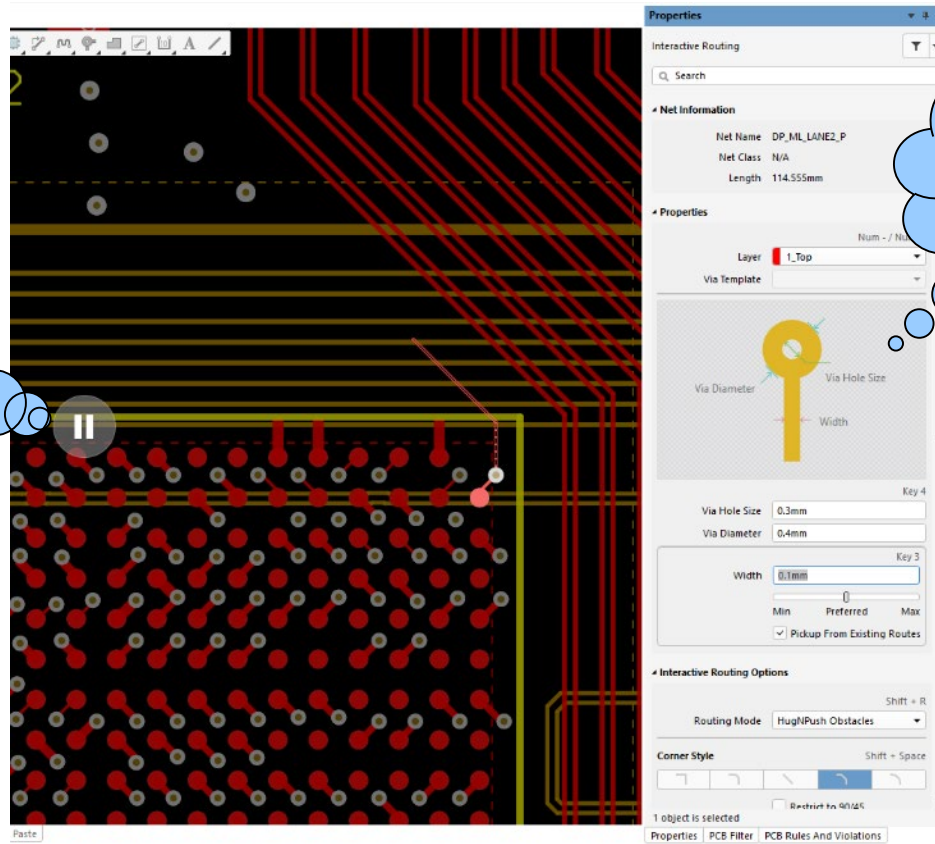
The board shape, also referred to as the board outline, is a closed polygon that defines the boundary, or extents, of the board.

This board outline provide a visual guideline of the extents of the space available for the placement and routing of the design, the board shape is also used by the internal power planes as the reference for the power plane edge pullback and the edge for splitting power planes.



Interactive Routing - Basics

Press Tab in
interactive
routing to
access the
Properties
panel



Adjust width,
via, corner
style, etc.

Interactive Routing - Vias

- Altium Designer provides you with ability to add vias on-the-fly during interactive routing
 - Vias can be added in only valid locations
 - The software will prevent you from placing vias if they conflict with objects on any of the layers
 - The properties of the via are determined by the applicable Routing Via Style design rule in the *PCB Rules and Constraints Editor* dialog
- Adding a Via On Layer Change
 - Press the * (asterisk) or + (plus) key on the numeric keypad while routing to insert a via and switch the routing to the next signal layer
 - Press the - (minus) key on the numeric keypad to insert a via and switch the routing to the previous signal layer
- Adding a Via Without Layer Change
 - Press the 2 key while routing to insert a via, however, keep routing on the current signal layer. Click to commit the via position

Via

- A via is a primitive design object
 - Vias are usually placed while routing, although in many cases it is interesting to place free vias (e.g RF circuits)
- Via properties allow for specifying same parameters for all layers (simple) or different
- Also allow for specifying blind and buried vias (start and end layer)
- Solder mask can be defined to open or tent vias
 - Negative expansions covers annular ring or complete tents via
- Vias can be declared as test points

How to define you own via templates: Create “Pad Via Library”, then open “PCB Pad Via template” panel in PCB editor, add via to internal library. Then you can use it in the design!

The screenshot shows the 'Properties' panel for a 'Via' object. The panel is organized into several sections:

- Via**: Contains a search bar and a dropdown menu.
- Net**: Includes a 'Net' dropdown set to 'GND', 'Net Class' set to 'N/A', and 'Net Length' set to '2926.319mm'.
- Via Template**: Includes a 'Template' dropdown set to 'v40h20m0mx0' and a 'Library' dropdown set to '<Local>'.
- Location**: Includes 'X/Y' coordinates set to '94.7mm' and '94.4mm', and a lock icon.
- Properties**: Includes a 'Drill Pair' dropdown set to '1_Top - 16_Bottom' and a 'Drill Pairs' button.
- Hole information**: Includes 'Hole Size' set to '0.2mm' and 'Tolerance' set to '+ N/A' and '- N/A'.
- Size and Shape**: Includes tabs for 'Simple', 'Top-Middle-Bottom', and 'Full Stack', and a 'Diameter' input set to '0.4mm'.
- Solder Mask Expansion**: Includes tabs for 'Rule' and 'Manual', and checkboxes for 'Tented' (checked) for both 'Top' and 'Bottom' layers, with a 'From Hole Edge' checkbox.



Agenda

Congrats!

**You've made it to the end of
the module**



Agenda

Congrats!

**You've made it to the end of
the module**